149.3 Complete native kernels and the exact condition for cancellation

149.3.1 The original augmented realization

Retain the original fine field U as a latent variable. Insert each native reference probability kernel in the old slots supplied by S58b, before action completion or connected expansion. Keep every original selector alternative. Integrating the new fiber and summing its alternatives gives one conditional on the **entire** retained old state. Successive insertions can therefore be realized by a joint auxiliary kernel

$$\mathsf{K}_U(d\zeta) = \sum_{\omega} \mathsf{K}_{U,\omega}(d\zeta), \qquad \int \mathsf{K}_U(d\zeta) = 1. \tag{WC11.11}$$

The history label ω includes the original component and forest choices with their actual weights. An individual selected $\mathsf{K}_{U,\omega}$ need not have unit mass. All old physical source evaluations remain in their old numerator slots. Coordinate changes may change how those slots are displayed; they do not replace their physical readout by the independent new reference output.

WC11.11 concerns the unexpanded finite augmented integral. It does not assert that an individual effective RG operation is a newly normalized probability kernel, or that a connected activity may be normalized separately. Such a change would replace the actual denominator and scalar record.

149.3.2 Full kernel intertwining

Apply WC11.5 to the integrated carrying function $\mathsf{K}\Psi$. Equivalently, at the joint level,

$$\rho e^{A}\mathsf{K} = e^{\mathcal{D}}\left(\rho e^{\mathcal{B}_{\mathcal{V}}(A)} e^{\mathcal{V}}\mathsf{K}\right), \tag{WC11.12}$$

and hence

$$\int \rho e^{A}\, \mathsf{K}\Psi = \int \rho e^{\mathcal{B}_{\mathcal{V}}(A)} (e^{\mathcal{V}}\mathsf{K})(e^{\mathcal{V}}\Psi). \tag{WC11.13}$$

Here the two factors on the right denote the full joint weak Leibniz expansion. Its unambiguous definition is $e^{\mathcal{V}}(\mathsf{K}\Psi)$. Derivatives act on original selectors, boundary distributions, constraints, coarea

and tree factors, effective scores, and live reference responses. They act on the original denominator as well.

On a matched fixed fiber where $\mathsf{K} = N/D$, the algebraic identity is literally

$$e^{\mathcal{V}}\mathsf{K} = \frac{e^{\mathcal{V}}N}{e^{\mathcal{V}}D}. \tag{WC11.14}$$

The denominator on the right is the pullback of the genuine denominator; it is not a freely chosen normalization. Moving-fiber or boundary terms are included by using the complete joint distribution in WC11.12. These equalities supply no quantitative inverse-denominator estimate on a complex mixed domain.

Normalization gives the exact coefficientwise identity

$$(e^{\mathcal{V}}\mathsf{K})1 = 1. \tag{WC11.15}$$

For an old physical carrying function $F(U)$, independent of the auxiliary variables, WC11.13 consequently becomes exactly WC11.5. Every auxiliary kernel response cancels in the complete selector sum and complete new-fiber integral. The intrinsic action B in WC11.6 remains.

This extends the closed-normalization statement NR10.5 to all finite current words with their full observable response. It is an equality before total variation, with no component-count estimate and no reserve expenditure. It is not a bound on the terms being cancelled.

149.3.3 The residual when the carrying function is not closed

For any function $\psi(U)$ independent of the new variables,

$$e^{\mathcal{V}}(\mathsf{K}\Psi) = e^{\mathcal{V}}\psi + (e^{\mathcal{V}}\mathsf{K})e^{\mathcal{V}}(\Psi - \psi). \tag{WC11.16}$$

The second term is the complete residual. It vanishes under the auxiliary integral when $\Psi = \psi$, but not merely because Ψ is gauge invariant, bounded, or geometrically far from a selected component.

At first order, the kernel part can be written

$$V(\mathsf{K}\Psi) = \mathsf{K}(V\Psi) + (V\mathsf{K})(\Psi - \mathsf{K}\Psi). \tag{WC11.17}$$

The centered expression uses $V\mathsf{K}(1) = 0$; it retains the full fine dependence of Ψ. At higher orders all ordered kernel, observable, and readout derivatives in WC11.13 remain. A retained output $q(U, \zeta)$ gives $\Psi = F(q(U, \zeta))$, so even its first derivative contains $Dq\,V$. Its later derivatives include the actual nonlinear readout jets. Geometric separation of two final coarse labels does not imply that these derivatives vanish.

For a genuine ancestral current $D_{V_I}\rho$, tested against an old physical function, the exact cancellation criterion is

$$V_I F = 0. \tag{WC11.18}$$

A sufficient criterion is that every actual original derivative anchor of V_I lies outside the determining set of F. The coefficient of V_I may have an enlarged determining set; it stays inside the complete

density adjoint. Enlarged coefficient dependence alone does not invalidate WC11.18. The issue is whether the **whole descendant packet**, including all contacts and kernel terms, is still the exact transported ancestral current.

To apply this criterion to a proposed packet in the construction, one must check all three facts on its original augmented realization:

- every selector alternative in that packet's conditional insertion is summed with its actual weight;
- every new auxiliary variable on which its external carrying function depends is either retained in the carrying function or integrated together with that complete dependence;
- after all compensation factors are multiplied back, the packet is the full derivative of the original joint density, with the old physical carrying function in its original slot.

A current record with its later scalar contacts removed does not meet the third condition. A healed branch with unsummed selectors does not meet the first. A generated activity whose read set crosses a new reference fiber does not meet the second merely because its anchor lies outside the fiber.

These are exact tests, not additional assumptions declared true for all MT36 histories.

149.4 Actual Wilson primitive contacts at every fixed source order

149.4.1 Physical normalization

Work with $SU(N)$, $N \geq 2$, on the original four-dimensional lattice. Let

$$S_0(U) = \sum_p s_p(U), \qquad s_p = N - \Re \operatorname{tr} U_p, \qquad \rho = Z^{-1} e^{-\beta S_0}, \qquad \beta = c_\beta g^{-2}. \tag{WC11.19}$$

For a link b and orthonormal color index A, write $X_{b,A}$ for the left Haar derivative. The weight f_b is real, field independent, and the same for all colors at that link. Define

$$F_{b,A} = X_{b,A} S_0, \qquad W_f = \sum_{b,A} f_b F_{b,A} X_{b,A}, \qquad \lambda_a = c_\beta^{-1} a^{-2},$$

$$V_f = \lambda_a W_f, \qquad t_f = \lambda_a \left(\beta \sum_{b,A} f_b F_{b,A}^2 - \sum_{b,A} f_b X_{b,A}^2 S_0 \right). \tag{WC11.20}$$

These are exactly the primitive physical current and vector field of PC10, since $\lambda_a = \beta^{-1} g^{-2} a^{-2}$. There is no additional a^4 in the physical source coefficient t_f.

Let $C_F > 0$ be the Casimir fixed by this generator normalization. The weighted Haar identity is

$$\Delta_f S_0 := \sum_{b,A} f_b X_{b,A}^2 S_0 = C_F \sum_p f_p (N - s_p), \qquad f_p = \sum_{b \in \partial p} f_b. \tag{WC11.21}$$

In particular, differentiating the weighted Laplacian produces a weighted Wilson force, not an uncontrolled constant term.

149.4.2 Connected microscopic support

Put an edge between two links if they belong to a common plaquette. Each force $F_{b,A}$ reads only the plaquette star of b. Expand a term in WC11.8 down to its individual derivative anchors $b_1, \dots, b_r$. A nonzero ordered term has the following property: after ordering its anchors in the order in which the derivatives are applied, each new anchor belongs to the union of the preceding plaquette stars. Thus the anchors form a connected set in the link interaction graph, allowing repetitions.

Proof. The last current in WC11.8 is a sum of terms supported on one plaquette star. A derivative outside the determining set of such a term gives zero. A nonzero application of W_f differentiates at a link in that set and appends only the force at that same link, whose determining set is its plaquette star. Induction proves the assertion. The constant part of WC11.21 is killed by the first derivative. □

Consequently B_I vanishes if its source-support graph is disconnected, where two label supports are adjacent when some pair of their links shares a plaquette. This is an exact statement about the primitive contact coefficient before taking its Wilson expectation. The expectation of products of two separated contact clusters can still contain real Wilson interactions. Connected source support of a single coefficient does not imply factorization of the physical measure.

149.4.3 Nonlinear compact energy bound

Let Y_I contain the plaquettes read by the connected anchor clusters just described, and let

$$E_{Y_I} = \beta \sum_{p \in Y_I} s_p.$$

For each fixed $r = |I| \geq 2$, there is a constant $C_{r,N}$, independent of the mesh, number of links, and $\beta > 0$, such that

$$\boxed{|B_I(U)| \leq C_{r,N} c_\beta^{-r} (1 + \beta^{-1}) a^{-2r} \left(\prod_{i \in I} \|f_i\|_\infty \right) E_{Y_I}(U).} \tag{WC11.22}$$

This holds on real compact fields. The carrier can conservatively be enlarged to the full union of the relevant fixed-order plaquette stars; the estimate never replaces the original constraints or selectors by a different support.

Proof. For an elementary plaquette derivative,

$$|X_{b,A} s_p| \leq C_N s_p^{1/2}, \qquad |X_{b_1,A_1} \cdots X_{b_k,A_k} s_p| \leq C_{k,N} \quad (k \geq 2). \tag{WC11.23}$$

The first estimate follows from $\|U_p - I\|_{\mathrm{HS}}^2 = 2 s_p$, tracelessness of the generator, and unitary conjugation. The other estimates follow directly by differentiating a fixed plaquette word on the compact group.

The first term in parentheses in WC11.20 is a sum of products of two elementary Wilson forces. Applying any W_f preserves this property: differentiating a force can remove one first derivative, but the prefactor $F_{b,A}$ supplied by W_f supplies another. Derivatives of the remaining bounded local coefficients add further bounded factors. Any extra first derivative is also bounded on the compact group.

For the second term, WC11.21 shows that its first W_f derivative is already a sum of products of two elementary Wilson forces. Subsequent derivatives preserve the same property. Therefore every term in the contact coefficient has at least two such factors, with a bounded local coefficient. Use

$$|(X_i s_p)(X_j s_q)| \le C_N(s_p + s_q).$$

At fixed r, bounded plaquette incidence gives only $C_{r,N}$ ordered clusters meeting a given plaquette once the source suprema are taken. Thus the sum of absolute values is bounded by

$$C_{r,N}\lambda_a^r(\beta+1)\left(\prod_i \|f_i\|_\infty\right)\sum_{p\in Y_I} s_p.$$

Substituting the definitions of λ_a and E_{Y_I} gives WC11.22. □

The estimate uses the one original Wilson energy. For several contact factors, their product can be paid against one retained reserve through the single inequality

$$E^k e^{-\delta E} \le \left(\frac{k}{e\delta}\right)^k,$$

with the actual total E and one δ. This does not supply any missing mesh power or a fresh reserve per label. The explicit a^{-2r} in WC11.22 remains.

If an incoming field-dependent coefficient multiplies a primitive current, the Ward derivatives also act on that coefficient. WC11.22 does not bound those extra jets. Their actual mixed-domain estimates remain part of LM47/MT36.

149.4.4 Higher contact order can stay quadratic in the field

For the uniform weight $f = 1$, let x be ordinary link coordinates near the flat field and let H be the Hessian of the dimensionless Wilson action there:

$$S_0(e^x) = \tfrac{1}{2}x^T H x + O(x^3).$$

The exact identity $\Delta S_0 = 4C_F(N\#p - S_0)$ gives

$$(t_1)_{[2]} = \lambda_a\big(\beta x^T H^2 x + 2C_F x^T H x\big), \qquad (W_1)_{[1]} = (Hx)\cdot\nabla.$$

For r distinct labels carrying this identical primitive current, WC11.8 gives $B_I = (r-1)V_1^{r-1}t_1$. Its quadratic Taylor coefficient is consequently

$$\boxed{(B_I)_{[2]} = (r-1)2^{r-1}\lambda_a^r\big(\beta x^T H^{r+1}x + 2C_F x^T H^r x\big).} \qquad \text{(WC11.24)}$$

It is nonzero on positive Hessian directions. This is the Taylor coefficient of the exact nonlinear Wilson formula. It is not a replacement of the measure by a Gaussian law. It directly rules out the inference that greater current-derivative order by itself raises the first nonzero field degree.

149.5 The genuine second-source scalar coefficient

For two distinct labels carrying the same real primitive V, let

$$\mathcal{V} = (\eta_1 + \eta_2)V, \qquad t = (\eta_1 + \eta_2)t_V, \qquad t_V = \rho^{-1} D_V \rho.$$

The exact scalar identities are

$$\mathbb{E}_\mu t_V = 0, \qquad \mathbb{E}_\mu(V t_V) = \mathbb{E}_\mu(t_V^2), \qquad B_{\{1,2\}} = V t_V. \tag{WC11.25}$$

The middle equality is WC11.1 with test t_V. Hence

$$[\eta_1\eta_2] \log \mathbb{E}_\mu e^{(\eta_1+\eta_2)t_V} = \mathbb{E}_\mu(t_V^2) = \mathbb{E}_\mu B_{\{1,2\}} \geq 0. \tag{WC11.26}$$

This is a physical normalization statement for the original source exponential. The signed pointwise function $B_{\{1,2\}}$ need not be nonnegative. Its integrated scalar coefficient is nonnegative, and its absolute scalar mass satisfies

$$\int |B_{\{1,2\}}|\, d\mu \geq \mathbb{E}_\mu(t_V^2). \tag{WC11.27}$$

Complete auxiliary normalization leaves this number unchanged. A scalar cancellation can occur if the **original** source prescription already contains correlated additional contact coefficients; it cannot be inferred from the fact that an auxiliary kernel has total mass one. Nothing here asserts that every generated D6 packet is independently exponentiated in the actual construction.

149.6 A uniform nonlinear lower bound for the physical D6 vacuum contact

This section strengthens WC11.26 for the exact Wilson measure. Fix the gauge group and the normalization c_β. Take a periodic isotropic lattice with $n = T/a$ sites in each direction. Assume n exceeds a fixed geometric constant so that the finite neighborhoods below do not wrap around the torus.

Let

$$G_\beta = \sum_{b,A} (X_{b,A} S_0)^2 - \beta^{-1} \Delta S_0, \qquad t_6 = g^{-2} a^{-2} G_\beta. \tag{WC11.28}$$

This is the uniform primitive current in WC11.20.

Theorem. There are constants $\beta_0, c > 0$, depending only on the fixed gauge group and normalization, such that for every $\beta \geq \beta_0$,

$$\boxed{\mathrm{Var}_{\mu_\beta}(t_6) \geq c\, T^4 a^{-8}.} \tag{WC11.29}$$

No continuum or weak-coupling asymptotic expansion is used in the proof. In particular the constant is uniform in the number of sites and in $\beta \geq \beta_0$.

149.6.1 Uniform local small-action probability

The elementary partition-function bound is

$$-\log Z(\beta) \le C\,\#\{\text{links}\}\log(2+\beta). \tag{WC11.30}$$

Indeed restrict every link to a Haar ball of radius $c\beta^{-1/2}$ around the identity. Its Haar volume is at least $c'\beta^{-\dim SU(N)/2}$. Every plaquette action is then at most $C\beta^{-1}$. Multiplying the volumes and the Boltzmann lower bound proves WC11.30, uniformly in the lattice size.

Monotonicity of $\mathbb{E}_\beta S_0$ in β gives

$$\frac{\beta}{2}\mathbb{E}_\beta S_0 \le \log Z(\beta/2) - \log Z(\beta) \le -\log Z(\beta).$$

Translation and lattice rotation invariance therefore imply

$$\mathbb{E}_\beta s_p \le C\frac{\log(2+\beta)}{\beta} \tag{WC11.31}$$

for each plaquette. For any fixed finite contractible plaquette patch Q and any fixed $\varepsilon > 0$,

$$\mu_\beta\{s_p \le \varepsilon^2 \text{ for every } p \in Q\} \ge 1 - C_Q\varepsilon^{-2}\frac{\log(2+\beta)}{\beta}. \tag{WC11.32}$$

Only a fixed patch is used; its size does not grow with the torus.

149.6.2 A positive-density set of conditionally independent free links

Use the link interaction graph from Section 4. Choose a set $\mathcal{F}$ of links whose mutual graph distances are at least five. A greedy packing gives

$$|\mathcal{F}| \ge c_0 n^4, \tag{WC11.33}$$

because that graph has uniformly bounded degree. Condition on all links outside $\mathcal{F}$, calling them the bath $\mathfrak{b}$.

No plaquette contains two free links. The conditional Wilson law therefore factors:

$$\mu_\beta(dU_{\mathcal{F}} \mid \mathfrak{b}) = \prod_{b\in\mathcal{F}} \mu_{b,\beta}(dU_b \mid \mathfrak{b}). \tag{WC11.34}$$

Every summand $(X_{i,A}S_0)^2$ reads only the plaquette star of its center link. Its determining set meets at most one free link under this separation. The Laplacian term is a sum of plaquette terms. Thus, with the bath fixed,

$$G_\beta = G_0(\mathfrak{b}) + \sum_{b\in\mathcal{F}} G_{b,\beta}(U_b;\mathfrak{b}). \tag{WC11.35}$$

The variable part $G_{b,\beta}$ reads a fixed finite neighborhood of b, contained in graph distance two. All its other links belong to the conditioned bath. Equations WC11.34–35 imply

$$\operatorname{Var}_{\mu_\beta}(G_\beta) \geq \sum_{b\in\mathcal{F}} \mathbb{E}_{\mu_\beta} \operatorname{Var}_{\mu_{b,\beta}}(G_{b,\beta} \mid \mathfrak{b}). \tag{WC11.36}$$

149.6.3 A good-bath event measurable in the conditioned links

Call the bath near b good if there exists a unitary vertex gauge for which all bath links in the finite determining neighborhood of $G_{b,\beta}$ lie within δ of the identity. This event depends only on conditioned links; the value of U_b is not part of its definition.

Choose a fixed contractible cube containing that neighborhood and the plaquettes needed to fill its tree cycles. If every plaquette of that cube has action at most ε^2, a spanning-tree gauge makes all its links lie within $C_Q\varepsilon$ of the identity. To see this, write each tree-cycle holonomy as a product of conjugates of a bounded number of plaquette holonomies. Unitarity and telescoping give

$$\|U_{\text{cycle}} - I\| \leq \sum_{p \text{ in its filling}} \|U_p - I\| \leq C_Q\varepsilon.$$

The gauge proving this implication may be chosen using the whole small-field configuration. This causes no measurability problem: the conclusion is the **existence** of a gauge putting the bath links close to the identity, which is a property of those bath links alone. Equivalently, the good-bath event can be defined by minimizing their maximum transformed distance over the compact finite vertex gauge group.

Take ε so that $C_Q\varepsilon < \delta$. Equation WC11.32 then gives, for all $\beta \geq \beta_0(\delta)$,

$$\mu_\beta(\mathfrak{b} \text{ good near } b) \geq \tfrac{1}{2}, \tag{WC11.37}$$

uniformly in the torus and in b. The small-plaquette event used to prove this may involve free-link values; the good-bath event itself does not. Only the latter is used in the conditional variance estimate.

149.6.4 The exact flat-bath calculation

Put every bath link in the determining neighborhood equal to the identity and vary only $U_b = U$. In four dimensions there are six plaquettes containing b. Each has action

$$s(U) = N - \Re \operatorname{tr} U.$$

The force at b is six times the one-plaquette force. The other eighteen links in those six plaquettes are distinct, and each carries one such force, up to a unitary adjoint or sign. Thus

$$\sum_{i,A} (X_{i,A} S_0)^2 = 54\,|\nabla s(U)|^2 + \text{a bath constant}.$$

The exact Casimir identity gives $\Delta S_0 = 4C_F(N\#p - S_0)$. Hence the variable part of G_β is exactly

$$G^{\text{flat}}_{b,\beta}(U) = 54|\nabla s(U)|^2 + 24C_F\beta^{-1}s(U) + \text{a constant}. \tag{WC11.38}$$

The Hessian of the first term is strictly positive at the identity. In a sufficiently small exponential chart around it, the Hessian of $G_{b,\beta}^{\mathrm{flat}}$ is bounded below by a positive constant, uniformly for $\beta \geq 1$.

All expressions involve only finitely many fixed plaquette words. By continuity in the bath links and in $\beta^{-1} \in [0,1]$, for a sufficiently small fixed δ the same positive Hessian bound holds on a smaller fixed chart for every good bath, after the gauge in its definition. Gauge invariance and Haar invariance preserve the conditional variance. This is a uniform finite-dimensional statement; no typical fluctuation expansion is invoked.

149.6.5 Uniform one-link density and variance bounds

For a good bath the conditional one-link action has the form

$$S_b(U;\mathfrak{b}) = 6N - \Re\,\mathrm{tr}(UH_b), \qquad \|H_b - 6I\| \leq C\delta, \tag{WC11.39}$$

up to terms independent of U. At $H_b = 6I$, this action has a unique minimum at $U = I$, a positive Hessian there, and a positive gap outside any fixed sufficiently small identity neighborhood. Compactness, the implicit function theorem, and the positive Hessian give uniformly for H_b in the indicated fixed neighborhood:

- a unique global minimum in the chart, with coordinate $x_*(\mathfrak{b})$;
- positive upper and lower Hessian bounds near that minimum;
- a positive action gap outside the chart.

It follows by direct integration that the one-link normalizing integral, after subtracting its minimum action, lies between $c\beta^{-d/2}$ and $C\beta^{-d/2}$, where $d = \dim SU(N)$. In the rescaled coordinates

$$z = \sqrt{\beta}\,(x - x_*),$$

the normalized conditional density is bounded below by a positive constant on a fixed unit Euclidean ball. This lower bound is uniform in $\beta \geq \beta_0$ and in every good bath.

A possible linear term in $G_{b,\beta}$ at the conditional minimum must not be ignored. The following argument handles it directly. Put

$$H(z) = \beta G_{b,\beta}(x_* + \beta^{-1/2} z;\mathfrak{b}).$$

By Section 6.4, $D_z^2 H \geq cI$ on that ball. Choose a nonnegative $\varphi \in C_c^\infty$ of the ball with $\int \varphi = 1$. For every constant m,

$$dc \leq \int (\Delta H)\varphi = \int (H - m)\Delta\varphi \leq \|H - m\|_{L^2}\,\|\Delta\varphi\|_{L^2}.$$

The uniform conditional density lower bound therefore yields

$$\mathrm{Var}_{\mu_{b,\beta}}(G_{b,\beta} \mid \mathfrak{b}) \geq c_1\beta^{-2} \quad \text{on every good bath.} \tag{WC11.40}$$

This proof is insensitive to a gradient at x_* that is large after $\sqrt{\beta}$ rescaling. It neither drops that gradient nor assumes a uniform Gaussian limit.

149.6.6 Completion of the lower bound

Equations WC11.33, WC11.36, WC11.37, and WC11.40 give

$$\mathrm{Var}_{\mu_\beta}(G_\beta) \geq c\beta^{-2}n^4.$$

Using WC11.28 and $\beta = c_\beta g^{-2}$,

$$\mathrm{Var}_{\mu_\beta}(t_6) = g^{-4}a^{-4}\,\mathrm{Var}_{\mu_\beta}(G_\beta) \geq cc_\beta^{-2}a^{-4}n^4 = c'T^4a^{-8}.$$

This proves WC11.29. In particular, for two distinct labels carrying this same primitive current,

$$\mathbb{E}_\mu B_{\{1,2\}} = \mathrm{Var}_{\mu_\beta}(t_6) \geq cT^4a^{-8}, \qquad \int |B_{\{1,2\}}|\,d\mu \geq cT^4a^{-8}. \tag{WC11.41}$$

The sharp absolute masses in WC11.41 and WC11.44 are taken against the stated Wilson probability μ_β. Against the raw measure $e^{-\beta S_0}d\lambda$, they are multiplied by its genuine mass $Z(\beta)$; an independent field-independent carried factor e^s multiplies them further by $|e^s|$. The physical logarithmic source coefficient remains $\mathrm{Var}_{\mu_\beta}(t_6)$. No inverse partition function is inserted into an unnormalized scalar record.

A true factor supplied by the original source normalization multiplies this bound by its true product. The theorem does not insert such a factor. It also does not prohibit an explicitly correlated scalar counterterm already present in a different physical source prescription.

149.6.7 A fixed physical insertion can carry the same scalar mass

The argument is not restricted to a source covering the entire torus. Let the real, field-independent link weight f equal one, in every direction, on a fixed physical box Q of positive volume. Its bounded values outside that box can be arbitrary. Choose the separated free links only in an interior sub-box with a microscopic collar large enough that every center whose force star touches a selected free link has its full determining neighborhood in Q. Every weighted force and plaquette term that depends on one of those free links then has exactly the uniform weight used in WC11.38.

The conditional factorization and additivity still hold, and the number of selected links is at least $c|Q|a^{-4}$ for sufficiently small a. The same proof gives

$$\mathrm{Var}_{\mu_\beta}(t_f) \geq c|Q|a^{-8}, \qquad \beta \geq \beta_0. \tag{WC11.42}$$

The constant depends on the chosen interior box and the fixed group normalization. No condition on the source weight outside its determining neighborhoods is used in the conditional variance lower bound. The result still concerns two distinct source labels carrying that primitive physical current; it does not assert that a generated effective packet has this uncancelled source prescription.

149.6.8 Matching upper bound for the uniform physical current

The exact variance has the same upper power. For the uniform weight write $F_i = X_iS_0$, $W = \sum_i F_iX_i$, and $\lambda_a = c_\beta^{-1}a^{-2}$. The complete Haar Hessian $H_{ij}(U) = X_iX_jS_0(U)$ has uniformly bounded row and

column sums: a mixed derivative vanishes unless the two links belong to a common plaquette, and every remaining fixed plaquette derivative is uniformly bounded on the compact group. Therefore

$$\left|W\sum_i F_i^2\right| = \left|2\sum_{i,j} F_i H_{ij} F_j\right| \le C\sum_i F_i^2.$$

This does not assume that the ordered Haar Hessian is symmetric. Its symmetric part has the same uniform operator bound.

Using the exact identity $\Delta S_0 = 4C_F(N\#p - S_0)$, WC11.25 gives

$$\mathbb{E}_\mu t_6^2 = \lambda_a^2 \mathbb{E}_\mu\left[\beta W\sum_i F_i^2 + 4C_F\sum_i F_i^2\right].$$

A single exact Haar integration by parts gives

$$\beta\mathbb{E}_\mu\sum_i F_i^2 = \mathbb{E}_\mu \Delta S_0 \le 4C_F N\#p.$$

The same pointwise Hessian bound also gives $|Vt_6| \le C\lambda_a^2(\beta+1)\sum_i F_i^2$. Consequently, for $\beta \ge 1$,

$$\mathbb{E}_{\mu_\beta}|Vt_6| \le C\lambda_a^2(1+\beta^{-1})\#p \le C'T^4a^{-8}. \tag{WC11.43}$$

Together with WC11.29 this determines the scalar coefficient's order in the exact nonlinear model:

$$cT^4a^{-8} \le [\eta_1\eta_2]\log \mathbb{E}_{\mu_\beta} e^{(\eta_1+\eta_2)t_6} = \mathbb{E}_{\mu_\beta} B_{\{1,2\}} \le \mathbb{E}_{\mu_\beta}|B_{\{1,2\}}| \le CT^4a^{-8}, \qquad \beta \ge \beta_0. \tag{WC11.44}$$

Both constants are uniform in the cutoff and in the indicated weak-coupling range. The signed integrated coefficient and the absolute mass are distinct quantities; the displayed inequalities prove that both have this order without equating them.

149.6.9 Absolute contact mass for a general bounded physical insertion

The absolute upper bound also holds for every real, field-independent, color-independent link weight f. Let N_f be the number of links where $f \ne 0$. Bounded Hessian incidence gives

$$\left|W_f\sum_j f_j F_j^2\right| \le C\|f\|_\infty^2 \sum_{i:\, f_i\ne 0} F_i^2.$$

For the other term, bounded local third Haar derivatives give $|X_i\Delta_f S_0| \le C\|f\|_\infty$, and hence

$$|W_f\Delta_f S_0| \le C\|f\|_\infty^2 \sum_{i:\, f_i\ne 0}|F_i|.$$

The exact one-direction integration by parts implies $\mathbb{E}F_i^2 = \beta^{-1}\mathbb{E}X_i^2 S_0 \le C\beta^{-1}$, so $\mathbb{E}|F_i| \le C\beta^{-1/2}$. With the fixed color multiplicity absorbed in the constant, these estimates prove

$$\boxed{\mathbb{E}_{\mu_\beta}|V_f t_f| \le C c_\beta^{-2} a^{-4} \|f\|_\infty^2 N_f (1+\beta^{-1/2}).} \tag{WC11.44a}$$

Thus a bounded physical test supported in a fixed region has an $O(a^{-8})$ absolute second-contact mass for $\beta \ge 1$. If it equals one on a fixed positive-volume box as in WC11.42, then for sufficiently small a,

$$c_f a^{-8} \le \mathbb{E}_{\mu_\beta} B_{\{1,2\}} = \mathrm{Var}_{\mu_\beta}(t_f) \le \mathbb{E}_{\mu_\beta}|B_{\{1,2\}}| \le C_f a^{-8}, \qquad \beta \ge \beta_0, \tag{WC11.44b}$$

where both labels carry this same primitive physical test. The constants depend on its plateau, support volume, and supremum. The estimate uses only the original source normalization. Correlated companions of a generated packet must still be included before applying this primitive-source conclusion to that packet.

150.1 The first-current cancellation on the original realization

Let μ be a positive finite measure on the actual augmented configuration space. Its total mass is retained. Suppose a smooth vector field V and an ordinary scalar t obey the complete integration-by-parts identity

$$\int tF\,d\mu = \int VF\,d\mu. \tag{SC11.1}$$

All actual constraint and reference components of V are included. For the moment SC11.1 has no omitted boundary functional. A selected history for which that is not true is treated in Section 4.

Let q be the actual smooth observation on this realization, including every live reference coordinate that its directions vary. Write

$$\rho = q_*\mu, \qquad \bar{h} = \mathbb{E}(h \mid q), \qquad v = Dq\,V, \quad \bar{v} = \mathbb{E}(v \mid q), \quad \bar{t} = \mathbb{E}(t \mid q). \tag{SC11.2}$$

The conditional laws are probabilities, but ρ carries the original absolute scalar mass. No rank-deficient observation is declared nonsingular. All formulas take place on an identified smooth support chart of its actual range; when no such chart is available, the preceding tested identity, not an invented Lebesgue density, is the definition.

Use $D = -\operatorname{div}$, so $\langle D\nu, \varphi\rangle = \int \nabla\varphi \cdot d\nu$. The chain rule gives $\bar{t}\rho = D(\bar{v}\rho)$. In the derivative formulas below, assume that the actual conditional mean $\bar{h}$ is differentiable to the displayed order and that the required products are valid integrable tests of the pushed identity. Smoothness of q and h alone does not prove this property. Section 5 gives the exact formulas on a legitimate differentiable conditional chart; it does not assert such a chart for every history. For a smooth multiplier h with this regularity, set

$$k_t = \operatorname{Cov}(h, t \mid q), \qquad k_v = \operatorname{Cov}(h, v \mid q).$$

Then the entire covariance packet satisfies

$$\boxed{k_t\rho - D(k_v\rho) = \left[\mathbb{E}(Vh \mid q) - \bar{v}\cdot\nabla\bar{h}\right]\rho.} \tag{SC11.3}$$

This is an ordinary scalar representation of the *difference* on the left. To prove it, apply SC11.1 to $h\varphi\circ q$:

$$q_*(ht\mu) = D(q_*(hv\mu)) + q_*((Vh)\mu).$$

Apply the already pushed identity to $\bar{h}\varphi$, and subtract. Both fine and retained normalizations are still present.

In particular,

$$\mathfrak{M}(k_t\rho - D(k_v\rho)) \le \int \left|\mathbb{E}(Vh \mid q) - \bar{v}\cdot\nabla\bar{h}\right| d\rho. \tag{SC11.4}$$

Here $\mathfrak{M}$ is the represented-distribution norm of CT9.3, allowing the empty word. This is not a bound for $\int |k_t| d\rho$ by itself. The right side uses actual fine and retained derivatives. A small supremum of $h-1$ in an outer parameter does not control it.

If $h = H\circ q$, both sides vanish exactly. More generally, subtracting any such $H\circ q$ from h leaves the packet unchanged. Thus its dependence is on the part of the multiplier that is not a function of the complete retained observation. A proof that an inherited interaction is almost a retained function must control this difference in the stated derivative norms, not only pointwise.

150.2 Second currents and the lower current they necessarily create

Let a differential operator of order at most two, with no zero-order term, have the form

$$L = \sum_{i,j} H_{ij}X_jX_i + \sum_i B_iX_i, \qquad \int tF\,d\mu = \int LF\,d\mu. \tag{SC11.5}$$

The vector fields, their order, and all physical coefficients are literal. Neither the X_i nor the matrix H is assumed constant or commuting. Coefficients stand to the left of the ordered word. Define

$$\begin{aligned} a^{\alpha\beta} &= \sum_{i,j} H_{ij}(X_jq^\alpha)(X_iq^\beta),\\ b^\alpha &= \sum_{i,j} H_{ij}X_jX_iq^\alpha + \sum_i B_iX_iq^\alpha,\\ c_h^\alpha &= \sum_{i,j} H_{ij}\left[(X_jh)(X_iq^\alpha) + (X_ih)(X_jq^\alpha)\right]. \end{aligned} \tag{SC11.6}$$

Thus $L(\varphi\circ q) = a^{\alpha\beta}\partial_\alpha\partial_\beta\varphi + b^\alpha\partial_\alpha\varphi$, evaluated at q. The complete product rule is

$$L(h\varphi\circ q) = hL(\varphi\circ q) + (Lh)\varphi\circ q + c_h^\alpha\partial_\alpha\varphi\circ q. \tag{SC11.7}$$

This identity includes derivatives of the field-dependent frames inside each ordered word.

Use bars for conditional means and put

$$\bar{L} = \bar{a}^{\alpha\beta}\partial_\alpha\partial_\beta + \bar{b}^\alpha\partial_\alpha, \qquad \bar{c}_H^\alpha = (\bar{a}^{\alpha\beta} + \bar{a}^{\beta\alpha})\partial_\beta H. \tag{SC11.8}$$

Let $k_a = \mathrm{Cov}(h, a \mid q)$ and $k_b = \mathrm{Cov}(h, b \mid q)$. Then

$$\boxed{\begin{aligned} k_t\rho - D_\alpha D_\beta(k_a^{\alpha\beta}\rho) - D_\alpha(k_b^\alpha\rho)&\\ = R_0\rho + D_\alpha(R_1^\alpha\rho),&\\ R_0 = \mathbb{E}(Lh \mid q) - \bar{L}\bar{h},&\\ R_1^\alpha = \mathbb{E}(c_h^\alpha \mid q) - \bar{c}_{\bar{h}}^\alpha.& \end{aligned}} \tag{SC11.9}$$

The positive sign before the last D is fixed by the convention $D = -\partial$. To verify SC11.9, push SC11.7 forward, perform the same product expansion with $\bar{L}$, and subtract. The two second-order terms

whose coefficients contain h give k_a; their first-order counterparts give k_b. No Hermitian conjugation is inserted in a complex coefficient.

An explicit resulting norm bound is

$$\mathfrak{M}(\text{left side of SC11.9}) \le \int \left(|R_0| + \sum_{\alpha} |R_1^{\alpha}| \right) d\rho. \tag{SC11.10}$$

This uses a represented norm whose allowed target words include the displayed D_α words. Their original canonical and source weights are already in R_1; if a different fixed word list is used, a justified conversion with its literal coefficients is needed. All original target-coordinate incidences are in the displayed sum. If $r = h - \bar{h} \circ q$ is smooth, the same coefficients may be written

$$R_0 = \mathbb{E}(Lr \mid q), \qquad R_1 = \mathbb{E}(c_r \mid q). \tag{SC11.11}$$

This follows because the full chain rule for $L(\bar{h} \circ q)$ is exactly $\bar{L}\bar{h}$ after conditioning. It makes the additional cancellation explicit while keeping the derivatives of the conditional mean. It does not show that those derivatives are small.

150.3 All finite ordered source words

The same argument works at any fixed finite order. Suppose $L = \sum_w A_w X_w$ has order at most r, no constant term, and the complete adjoint identity against μ. Expand its actual chain rule as

$$L(\varphi \circ q) = \sum_{1 \le |\alpha| \le r} c_\alpha (\partial^\alpha \varphi) \circ q. \tag{SC11.12}$$

This is a local-coordinate description; the original ordered fine words are not commuted to obtain it. Their brackets and all lower chain-rule terms are already in c_α. Define the product coefficients $r_{h,\alpha}$ by

$$L(h\varphi \circ q) - hL(\varphi \circ q) = \sum_{0 \le |\alpha| \le r-1} r_{h,\alpha} (\partial^\alpha \varphi) \circ q. \tag{SC11.13}$$

They exist by the ordinary ordered Leibniz rule. Every summand differentiates h at least once, and $r_{h,0} = Lh$. Let $\bar{L} = \sum \bar{c}_\alpha \partial^\alpha$, and let $\bar{r}_{H,\alpha}$ denote its analogous product coefficients. Then

$$\begin{aligned} &\operatorname{Cov}(h, t \mid q)\rho - \sum_{1 \le |\alpha| \le r} D^\alpha \big(\operatorname{Cov}(h, c_\alpha \mid q)\rho\big) \\ &\qquad = \sum_{0 \le |\alpha| \le r-1} D^\alpha \left(\left[\mathbb{E}(r_{h,\alpha} \mid q) - \bar{r}_{\bar{h},\alpha} \right] \rho \right). \end{aligned} \tag{SC11.14}$$

Testing against φ, using SC11.13 on the fine space and its retained counterpart, proves the assertion. In particular, the new scalar coefficient is always $\mathbb{E}(Lh \mid q) - \bar{L}\bar{h}$; it is not set to zero.

For the original finite commutative source algebra, apply SC11.14 coefficientwise after the actual ordered word has been formed. The Leibniz expansion preserves every label partition, test offset, original word mesh and birth coupling. No independent replica of a shared factor is introduced. This is a finite algebraic closure statement, not a uniform estimate in word order or history depth.

150.4 Selectors and the complete boundary remainder

SC11.1 or SC11.5 is not automatically true for the restricted measure of an individual history. If the literal identity is instead

$$t\mu = L^*\mu + \mathcal{B}, \tag{SC11.15}$$

the entire boundary/constraint distribution $\mathcal{B}$ must be retained. For smooth h and $\bar{h}$, the right side of SC11.14 acquires exactly

$$q_*(h\mathcal{B}) - \bar{h}\, q_*\mathcal{B}. \tag{SC11.16}$$

This follows by retaining the extra tested term in both fine and retained integrations by parts. It is an identity of already defined distributions, not a product of independently exposed delta currents.

If a multiplier itself is sharp, SC11.13 need not have ordinary coefficient derivatives. One must use CT9's measure representation and the complete original joint kernel, or establish the required distributional product on that realization. Smoothing an original selector is not part of this proof. A sum of complete selector alternatives can cancel SC11.16 only with the same carrying function and all original responses included. It does not cancel a separately selected history.

SC11.16 also records the obstruction to using an integration by parts on a different reference fiber. The original constraint rows, gauge pins, coarea factors, and live-reference lift must first identify the common realization. No change of those data is supplied by the conditional expectation notation.

150.5 Genuine denominator derivatives in the retained coefficients

The derivatives of $\bar{h}$ in SC11.3–14 are derivatives of an actual conditional expectation. On a legitimate common fiber coordinate chart, let its full unnormalized density be $R_y(\xi)$ and set

$$Z_y = \int R_y(\xi)d\xi, \qquad N_y = \int h(y,\xi)R_y(\xi)d\xi, \qquad \bar{h} = N_y/Z_y. \tag{SC11.17}$$

These are the original densities, including any coordinate Jacobian, constraints resolved in the chart, and independent scalar factors. Where differentiation is justified, the exact quotient identities are

$$\begin{aligned} \partial_\alpha \bar{h} &= Z_y^{-1}(\partial_\alpha N_y - \bar{h}\partial_\alpha Z_y), \\ \partial_\alpha\partial_\beta \bar{h} &= Z_y^{-1}(\partial_\alpha\partial_\beta N_y - \bar{h}\partial_\alpha\partial_\beta Z_y - (\partial_\alpha \bar{h})\partial_\beta Z_y - (\partial_\beta \bar{h})\partial_\alpha Z_y). \end{aligned} \tag{SC11.18}$$

The normalization commutators are explicit. No lower bound on a raw conditional denominator is inferred from a bound on a separately extracted reference integral.

On a fixed smooth positive support, write $s_\alpha = \partial_\alpha \log R_y$. Then SC11.18 gives

$$\begin{aligned} \partial_\alpha \bar{h} &= \mathbb{E}_y(\partial_\alpha h) + \mathrm{Cov}_y(h, s_\alpha), \\ \partial_\beta\partial_\alpha \bar{h} &= \mathbb{E}_y(\partial_\beta\partial_\alpha h) + \mathrm{Cov}_y(\partial_\alpha h, s_\beta) + \mathrm{Cov}_y(\partial_\beta h, s_\alpha) \\ &\quad + \mathrm{Cov}_y(h, \partial_\beta s_\alpha) + \kappa_y(h, s_\alpha, s_\beta). \end{aligned} \tag{SC11.19}$$

Here κ_y is the third centered cumulant. Its arguments are paired bilinearly, with no source conjugation. Differentiating the first line and expanding the two covariance derivatives proves the second. The score is the score of the *full* density R_y, including inherited effective interactions and live-reference dependence. It is not replaced by the bare Wilson score.

If a selector boundary moves with y, the derivatives of N_y and Z_y retain the same boundary distributions. Taking the logarithm of a sharp selector to define s_α is not allowed; SC11.18 is then the appropriate tested formulation, provided those derivatives have actually been constructed.

150.6 Application to the original nonlinear Wilson Hessian current

NH9.5 gives a concrete nonlinear operator to which the preceding identities apply. With the original word's a^4, put

$$\begin{aligned} L^{\mathrm{own}} &= 2c_\beta^{-3}\sum_{i,j}(X_iX_jS_0)Y_jY_i + \zeta c_\beta^{-3}\sum_i g_jX_iS_0\,Y_i, \\ Y_i &= g_jX_i, \qquad \zeta = 3C_A/2 - 16C_F. \end{aligned} \tag{SC11.20}$$

For the bare Wilson body $\mu = e^{-\beta S_0}d\lambda$, with its original absolute normalization, its scalar adjoint is exactly the whole current on the left of NH9.5, including the scalar companion. This identification is not made with an arbitrary inherited body. An inherited action changes the adjoint score, and a selected or augmented constrained realization needs its complete lift and boundary packet before SC11.15 is used. If the physical source lacks that a^4, the actual operator in SC11.5 is instead

$$L^{\mathrm{phys}} = a^{-4}L^{\mathrm{own}}. \tag{SC11.21}$$

All conditional covariance coefficients retain this factor. The cancellation does not create a replacement power of the mesh.

For a smooth, nonzero relative multiplier $h = e^W$ on the original fine realization, the exact scalar response is

$$h^{-1}L^{\mathrm{phys}}h = a^{-4}c_\beta^{-3}\Big[2\sum_{i,j}(X_iX_jS_0)(Y_jY_iW + (Y_jW)(Y_iW)) + \zeta\sum_i g_jX_iS_0\,Y_iW\Big]. \tag{SC11.22}$$

This is an exact same-Wilson nonlinear identity. The mixed Hessian and both ordered derivatives remain. For q with the original responses $A_i^\alpha = Y_iq^\alpha$, its additional first-current coefficient is

$$c_h^\alpha = 2a^{-4}c_\beta^{-3}h\sum_{i,j}(X_iX_jS_0)\big[(Y_jW)A_i^\alpha + (Y_iW)A_j^\alpha\big]. \tag{SC11.23}$$

The lower row in SC11.20 contributes to the scalar Lh, but not to this cross term. Applying SC11.9 subtracts the retained responses $\bar{L}\bar{h}$ and $\bar{c}_{\bar{h}}$, including SC11.18's true denominator terms, before taking any variation.

For example, a direct real bound for its fine scalar part is

$$\begin{aligned} \int|L^{\mathrm{phys}}h|d\mu \le a^{-4}c_\beta^{-3}\int|h|\Big[2\sum_{i,j}|X_iX_jS_0|\big(|Y_jY_iW| + |Y_jW|\,|Y_iW|\big) \\ +|\zeta|\sum_i|g_jX_iS_0|\,|Y_iW|\Big]d\mu. \end{aligned} \tag{SC11.24}$$

The analogous first-current bound follows directly from SC11.23. These are useful sufficient integrals because no large scalar $|t|$ occurs. They are not declared small: their internal density derivatives, original mesh factor, coordinate counts and boundary partners have not disappeared.

For the actual R component, NC bounds small changes in the admitted outer pair while keeping the integrated variables real and fixed. That statement does not bound the internal derivatives Y_iW and Y_jY_iW in SC11.22 on an arbitrary old marked readout. To apply NC here one must prove the actual chain from those fine directions into its outer chart, retain the reference response, and control all remaining fiber directions. AE8/LM47 is part of that missing identification. A fixed auxiliary reference tail estimate also does not identify the old fine current with that auxiliary variable.

150.7 What this closes and what it does not

SC11.3, SC11.9 and SC11.14 prove a nonlinear cancellation of the complete PF10 covariance packet. The proof works at every finite source order and displays every lower current. SC11.16 preserves the exact boundary defect; SC11.18–19 retain the effective scores and genuine conditional normalizers. SC11.20–24 instantiate the calculation with the actual nonlinear Wilson Hessian current and its physical normalization.

They leave a precise quantitative task: estimate the fiber-dependent part of the actual inherited multiplier in the contracted derivative norms of SC11.22–24, together with its conditional derivatives and SC11.16, on the same admitted mixed complex domain. Saved incoming scale gains and the single failed-sector reserve must multiply those literal estimates. Neither outer density smallness nor a regular prefix's high derivative order supplies them. Thus all-history MT36 is still unproved.

151.3 A weighted tail bound for the actual extracted variable

151.3.1 Precise finite-dimensional assumptions

On an admitted real reference primitive, suppose its original expansion has been verified in the form

$$f_G(B;b) = \exp\{-\tfrac{1}{2}\langle B, Q_G(b)B\rangle\}u_G(B;b), \qquad \lambda_M I \le Q_G(b) \le \Lambda_M I, \qquad e^{-C_M} \le u_G(B;b) \le e^{C_M}, \tag{RV11.4}$$

throughout its original cutoff. The bound includes the Haar Jacobian, the weighted linear term and the nonlinear remainder. The constants may depend on the fixed block parameter M and the gauge group, but not on the fine cutoff, the old real datum in this admitted class, or the current coupling. The native coordinate dimension is retained, with $n_G \le n_M$. Norm equivalence between link norms and Euclidean coordinates is included in these constants.

Assume that the cutoff contains the Euclidean unit ball in these free coordinates. More generally one can use any fixed inner ball and retain its radius in the following constants. For the admitted native primitive this inner-ball condition follows from RV11.2a for sufficiently small g_k, after the fixed norm equivalences. Simultaneously RV11.2b places its group variables inside the required fixed logarithm chart.

For the original admitted real R-II reference primitive, RV11.4 is the finite-dimensional content of the expansion and bounds just cited. Using it on a mixed entry still requires verifying that the actual fixed-target field and the full cutoff support are admitted there. The finite-dimensional argument below does not perform that verification.

151.3.2 The quantitative tail lemma

For every $p \ge 0$ and $0 \le \theta \le \lambda_M/4$,

$$\boxed{K_G\Big[1_{\{\|B\|_\infty > r\}}(1+|B|)^p e^{\theta|B|^2}\Big] \le C_{M,p}\exp(-\lambda_M r^2/16).} \tag{RV11.5}$$

The norm $\|B\|_\infty$ is the maximum actual free-link norm. After fixed norm equivalences it is at most the Euclidean norm $|B|$. No extension of the fixed-target map outside its original cutoff is used.

Proof. If v_n is the Euclidean unit-ball volume, then

$$\widehat{D}_G \geq z_{M,n} := v_n e^{-C_M - \Lambda_M/2} > 0. \tag{RV11.6}$$

Set $a = \lambda_M/2 - \theta \geq \lambda_M/4$. On $|B| > r$, split the Gaussian exponent into three parts:

$$(1+|B|)^p e^{-a|B|^2} \leq \underbrace{\sup_{t\geq 0}(1+t)^p e^{-at^2/4}}_{c_{p,a}<\infty} e^{-ar^2/4} e^{-a|B|^2/2}. \tag{RV11.7}$$

Integrating the last Gaussian on $\mathbb{R}^n$ only enlarges this positive upper bound; it does not extend the original nonlinear integrand. Division by RV11.6 gives RV11.5, with an explicit permissible constant

$$C_{M,p} \geq \max_{1\leq n\leq n_M} z_{M,n}^{-1} e^{C_M} c_{p,\lambda_M/4} (8\pi/\lambda_M)^{n/2}. \tag{RV11.8}$$

The zero-dimensional reference is the point mass and can be treated separately. This proves the claim.

The dependence on the polynomial degree is explicit. For $p \geq 1$,

$$c_{p,\lambda_M/4} \leq 2^{p-1}\left[1 + (8p/(e\lambda_M))^{p/2}\right]. \tag{RV11.8a}$$

Indeed $(1+t)^p \leq 2^{p-1}(1+t^p)$, and the maximum of $t^p e^{-\lambda_M t^2/16}$ occurs at $t^2 = 8p/\lambda_M$. Thus this is not a constant uniform in an unbounded contact degree. Its degree dependence must be carried in any graded induction; no saved scale gain is generated by the moment estimate.

151.3.3 Complete carrying functions and scores

Suppose an actual inherited carrier, after its scalar companions have been multiplied back together, satisfies on this same fiber

$$|F(B;b) - a(b)| \leq A(b)(1+|B|)^p e^{\theta|B|^2}. \tag{RV11.9}$$

Here a is independent of the newly integrated variable, but can depend on the complete old datum. RV11.5 gives a tail bound with factor $A(b)$. If $F = e^{s(b)}\widetilde{F}$, then the factor $e^{\operatorname{Re} s(b)}$ remains in $A(b)$. It cannot be discarded by centering or by probability normalization.

For actual fiber functions F, S with respective envelopes $A(1+|B|)^p e^{\theta|B|^2}$ and $B_0(1+|B|)^q e^{\eta|B|^2}$, RV11.5 bounds their product and their tail product whenever $\theta + \eta \leq \lambda_M/4$. If $2\theta, 2\eta \leq \lambda_M/4$, Cauchy–Schwarz also gives

$$|\operatorname{Cov}_{K_G}(F,S)| \leq \sqrt{\operatorname{Var}_{K_G} F \ \operatorname{Var}_{K_G} S} \leq C_{M,p,q} A B_0. \tag{RV11.10}$$

For complex-valued functions the variance in this inequality is the usual positive variance of the modulus; the covariance can be the bilinear covariance. The same estimate follows from its centered integral representation.

These are estimates for the literal functions on the literal auxiliary reference variable. An original fine-link current is not identified with B. A complete external score also differentiates the nonlinear map, reference, constraints and moving cutoff when they depend on that external parameter. Its envelope is an additional obligation. Likewise, RV11.10 supplies no positive mesh power: any inverse

mesh powers in A or B_0 survive. It does not supply the nonlinear conditional-covariance cancellation required by MT36.

If an envelope uses a previously allocated energy reserve, its exponent must fit the stated θ bound. This is a restriction on the existing single reserve. The estimate creates no new large-field charge or independent reserve.

151.4 Exact relative forest transport on a fixed physical record

151.4.1 What is fixed

Consider a finite connected link record, or a disjoint union of its original components. Fix its dynamic edges E, vertices V, pinned vertices P, boundary data, all determining rows and raw averaging prefixes, and its reference target V_Λ. Let G_0, G_1 be two spanning forests rooted at the same pins, with exactly one pin in each tree. Every change of gauge is required to equal the identity on P. In particular

$$q_{G_i} = |E| - |V| + |P|, \qquad n_{G_i} = m(|E| - |V| + |P|). \tag{RV11.11}$$

No pin, root or connected-component count has been suppressed. The statement applies only to forests actually allowed by the record.

Write the physical target as $W_e = C_e V_{\Lambda,e}$. The relative gauge action with the reference held fixed is

$$C_e \longmapsto C'_e = h_{x(e)} C_e V_{\Lambda,e} h_{y(e)}^{-1} V_{\Lambda,e}^{-1}. \tag{RV11.12}$$

Each pinned forest gives a unique representative with $C_e = 1$ on that forest. The original compact Haar change of variables has unit Jacobian. One way to verify this is to eliminate tree variables from the leaves inward. At each elimination the map is a left or right Haar translation and a unitary conjugation. The normalized gauge volume is one. Thus the two sections induce the same quotient Haar measure.

This does not yet identify two nonlinear Wilson integrands. For that, the complete fixed-target variational record must be equivariant under RV11.12, the actual branches must be admitted on the original supports, and the complete transported evaluated action must agree. Gauge covariance and uniqueness then identify their actions on their common physical domain. These hypotheses hold for a verified coordinate change of one physical record; they have not been checked for every pair of native preparatory records in S94.

151.4.2 A common core for the original small-link cutoffs

Let ℓ be the maximum number of edges from a vertex to its pinned root in either forest, and set $L_G = 2\ell + 1$. For real compact targets, the relative gauge transport satisfies

$$\max_e \|C'_e - 1\| \le L_G \max_e \|C_e - 1\|. \tag{RV11.13}$$

To prove this, solve the tree recursion $h_y = V_e^{-1} h_x C_e V_e$. Unitary conjugation preserves the norm, so $\|h_y - 1\| \le \|h_x - 1\| + \|C_e - 1\|$. Starting from $h = 1$ at a pin bounds h_x by its tree-path length times the maximum link deviation. Substituting this estimate in RV11.12 gives RV11.13. No flatness of V_Λ is needed for this algebraic estimate, but its compact reality is used.

For a fixed sufficiently small group logarithm chart, the exponential and logarithm norms are uniformly comparable. Consequently, if both native cutoffs are $\max_e |B_e| < R_{\chi,k}$, their intersection as subsets of the physical quotient contains the image of

$$\max_e |B_e| < R_{\chi,k}/\kappa_G \quad \text{in either section}, \qquad \kappa_G = 4(2\ell+1), \tag{RV11.14}$$

after a fixed adjustment for the chosen Lie-algebra norm. This applies when the actual group radius $g_k R_{\chi,k}$ lies in that chart. RV11.2b verifies that condition for the admitted native primitive at sufficiently small g_k. For the rectangular native component ℓ and n_G have bounds depending on M. If its complete record wraps, its injective physical lift and its actual paths must be checked before this finite-box statement is used in a torus comparison.

151.6 An unconditional Wilson check: the two-tree obstruction vanishes for a growing rescaled cutoff

The NR10 two-plaquette example admits a stronger positive conclusion than its fixed-radius Gaussian calculation. This conclusion uses its actual compact Wilson measure, without replacing that measure by its quadratic limit.

Let $G_{\mathrm{gauge}} = SU(N)$, $s(U) = 1 - N^{-1}\operatorname{Re}\operatorname{Tr} U$, and $w_g(U) = e^{-s(U)/g^2}$. In the first tree coordinates the complete quotient measure is $w_g(X)w_g(Z)\,dX\,dZ$; in the second tree the free variables are $(X, Y) = (X, ZX)$. This exact Haar change of variables is the one used in NR10.

Choose the original cutoff of that example to be a conjugation-invariant ball $\mathcal{B}_\eta$ around the identity, measured in a fixed local logarithm norm. Suppose

$$\eta(g) \longrightarrow 0, \qquad R(g) := \eta(g)/g \longrightarrow \infty. \tag{RV11.23}$$

Then the genuine two cutoff denominators satisfy

$$\boxed{|\log(D_1(g)/D_2(g))| \le C_N e^{-c_N R(g)^2}} \tag{RV11.24}$$

for sufficiently small g. In particular their ratio tends to one. The constants can be adjusted to include a fixed sufficiently small upper bound for η.

Proof. Let $d(U, 1)$ be the distance of a fixed bi-invariant Riemannian metric on the compact group; its square is smooth in a fixed identity chart. The Wilson one-plaquette action has its only zero at the identity and a positive quadratic Hessian there. Compactness and local norm equivalence therefore give $c\,d(U,1)^2 \le s(U) \le C\,d(U,1)^2$ in a fixed identity chart, with a positive action gap off that chart. Haar density in that chart is bounded above and below. Integration on a ball of radius g gives $\int w_g\,dU \ge c_N g^m$, where $m = N^2 - 1$. For a sufficiently small fixed $a > 0$, integration in the same chart and use of the off-chart action gap yield

$$\frac{\int e^{ad(U,1)^2/g^2} w_g(U)\,dU}{\int w_g(U)\,dU} \le C_N. \tag{RV11.25}$$

The off-chart numerator is exponentially small in g^{-2} and absorbs its factor g^{-m}. Markov's inequality gives $\mathbb{P}_g[d(U,1) > rg] \le C_N e^{-c_N r^2}$.

Under the complete quotient probability law, X, Z are independent with this one-plaquette distribution. If both lie in a sufficiently small fixed fraction of $\mathcal{B}_\eta$, then X, Z, ZX all lie in $\mathcal{B}_\eta$. Thus both native two-tree cutoff events have complements of probability at most $C_N e^{-c_N R(g)^2}$. Comparing each cutoff mass with the same complete quotient mass proves RV11.24, or one may apply RV11.17 to their intersection. No uniform Taylor estimate of order $gR(g)^3$ is needed. The exact Wilson energy supplies the tail.

This resolves the fixed-radius obstruction for this explicit Wilson example under the stated growing-cutoff hypotheses. The original R-operation cutoff independently satisfies these two radius limits by the verified law RV11.2a–b. Neither fact identifies its native fixed-target records with this graph. The argument supplies no physical torus-distance factor.

151.7 Quantitative analytic propagation on a common original chart

151.7.1 A dimension-independent propagation lemma

Let $E_{\mathbb{C}}$ be the complexification of a real normed space $E_{\mathbb{R}}$, with isometric complex conjugation. Thus $\|\operatorname{Re} z\|, \|\operatorname{Im} z\| \le \|z\|$. Center the ball at a real datum and write it as B_R after translation. Let $f : B_R \to \mathcal{A}$ be holomorphic, with values in a complex Banach space, and suppose

$$\sup_{z \in B_R} \|f(z)\| \le B, \qquad \sup_{x \in B_R \cap E_{\mathbb{R}}} \|f(x)\| \le \delta, \qquad 0 \le \delta \le B. \tag{RV11.26}$$

Then

$$\boxed{\sup_{z \in B_{R/8}} \|f(z)\| \le B^{1/3} \delta^{2/3}.} \tag{RV11.27}$$

There is no factor depending on $\dim E$. In particular the argument does not propagate the estimate through one parameter at a time.

Proof. Fix $z = x + iy \in B_{R/8}$. The case $y = 0$ is immediate. Otherwise put $v = y/\|y\|$, a real unit vector, and $\rho = R/2$. For every norm-one functional $\ell \in \mathcal{A}^*$, the function

$$g(\zeta) = \ell(f(x + \zeta v)), \qquad |\zeta| < \rho, \tag{RV11.28}$$

is holomorphic on a neighborhood of the closed disc. Indeed its arguments have norm at most $R/8 + R/2 = 5R/8 < R$. Its modulus is at most B on the upper semicircle and at most δ on the real diameter.

On the upper half-disc the harmonic function equal to one on the diameter and zero on the semicircle is

$$H(\zeta) = 1 - \frac{2}{\pi} \arg \frac{\rho + \zeta}{\rho - \zeta}. \tag{RV11.29}$$

The quotient maps that half-disc into the first quadrant, so the displayed argument is a single harmonic branch. The maximum principle applied to $\log |g|$ gives $|g(\zeta)| \le \delta^{H(\zeta)} B^{1-H(\zeta)}$. Zeros of g are handled

by the usual subharmonic truncation; the two diameter endpoints have zero harmonic measure. At $\zeta = i\|y\|$,

$$H(i\|y\|) = 1 - \frac{4}{\pi}\arctan(2\|y\|/R) \geq 1 - \frac{4}{\pi}\arctan(1/4) > \frac{2}{3}. \tag{RV11.30}$$

Since $\delta \leq B$, this proves $|\ell(f(z))| \leq B^{1/3}\delta^{2/3}$. Taking the supremum over the Banach dual proves RV11.27. If $\delta = 0$, the one-variable identity theorem on each such line proves the same conclusion; if $B = 0$ it is immediate.

151.7.2 Application to the native denominator difference

Assume that two original real reference primitives satisfy Sections 4–5 **uniformly for every real datum in the real slice of the same specified ball** B_R. Assume also that their original fixed-target and relaxed constructions have been identified on this common complex outer chart, and that the original analytic logarithms in RV11.2 exist there. These assumptions include all target rows, spatial weights, gauges, retained records and actual parameter margins. A real-point comparison alone is insufficient for RV11.26. The running coupling is held fixed and positive on each outer-field chart; $g_k \downarrow 0$ indexes the family of these estimates. Analytic dependence of a moving sharp cutoff on a coupling parameter is not assumed.

For a verified coordinate-forest change of the same complete physical record, suppose the classical and coordinate prefactors agree on that real slice. Their explicitly extracted logarithms are holomorphic, so equality on the real slice implies equality throughout the ball. One may prove this on the same real-direction complex lines, or by the Taylor coefficients at a real interior point. The genuine denominator difference is then

$$f(z) = \log D_1(z) - \log D_0(z) = I_1(z) - I_0(z). \tag{RV11.31}$$

The branches are the original real-positive logarithms continued on the declared common chart. Each $D_i = P_i e^{I_i}$ is nonzero there by this original representation. No zero-free domain larger than that chart is asserted.

R-II (1.10)–(1.11), printed p.358, supplies $|I_i(z)| \leq CM^5$ on its original admitted analytic domain. When both primitives have the declared common domain, it therefore gives the complex bound $B = 2CM^5$ for RV11.31. The real core comparison and the verified cutoff law give $\delta = C_M e^{-c_M p_0(g_k)^2}$, after enlarging the constants in RV11.17. For sufficiently small g_k, $\delta \leq B$, and RV11.27 proves

$$\boxed{\sup_{z \in B_{R/8}} |\log D_1(z) - \log D_0(z)| \leq C'_M e^{-c'_M p_0(g_k)^2}.} \tag{RV11.32}$$

For example one may take $c'_M = 2c_M/3$, absorbing the other fixed factors into C'_M. This proof compares the two holomorphic expressions on their original native cycles. It requires no identification of a transformed complex cycle with a real Haar cycle.

If the classical prefactors do not agree, the function to which the lemma is applied remains $f = \Delta \log P + \Delta I$. Its **entire** complex bound must then be supplied in RV11.26. A bound for ΔI alone does not bound f. Independently retained vacuum masses Δs likewise remain outside this conclusion unless they are explicitly included in f and both hypotheses of RV11.26 are proved for the complete expression.

Thus RV11.32 closes the auxiliary complex forest-cutoff comparison on a verified common native chart. It does not construct that chart for actual mixed histories or show that two native preparations alter only their forests.

151.7.3 Literal parameter radii and complete carriers

Let a parameter polydisc centered at z_0, with radii $d_1, \dots, d_n$, be contained in $B_{R/8}$. Cauchy's formula applied to RV11.27 gives

$$\|\partial^\nu f(z_0)\| \le \nu!\, B^{1/3}\delta^{2/3} \prod_{j=1}^{n} d_j^{-\nu_j}. \tag{RV11.33}$$

These are the actual simultaneous parameter radii. In a general normed ball one cannot assign a radius to every coordinate separately without verifying containment of the full polydisc. A shrinking original wall margin remains in RV11.33 and can consume a previously saved gain.

The Banach-valued lemma also applies to a complete carrier difference $f(z) = K_1(z)F_1(z) - K_0(z)F_0(z)$, provided the two complete normalized expressions are holomorphic on the same original chart, have the asserted complex bound, and satisfy the uniform real comparison RV11.19–20. Inherited scalar masses, reference and observable responses and all normalization commutators belong to those expressions. The bound on the logarithm of a bare reference denominator does not by itself establish these carrier hypotheses.

If a literal incoming factor $\Gamma \ge 0$, fixed on the declared chart, occurs in both bounds, say $B = \Gamma B_0$ and $\delta = \Gamma\delta_0$, then

$$B^{1/3}\delta^{2/3} = \Gamma B_0^{1/3}\delta_0^{2/3}. \tag{RV11.34}$$

Consequently a saved mesh or coupling factor common to the real and complex estimates is retained exactly. A real-only gain is not automatically retained with the same exponent. Neither statement produces a gain absent from the physical source. All boundary and constraint currents still need their own identified holomorphic representations before this lemma can be applied to them.

154.1 Result and limits

The actual numerator preparation has a protected sector that can be identified without replacing it by the relaxed minimum. It follows from the literal constraint statement immediately after R-II (1.50), printed p.370: the old and new determining sets, and their target field V'', coincide outside the active domain Λ. Consequently, an old carrying function which factors through those unchanged raw rows has exactly the same value on the old numerator minimum and the new minimum, including at nonzero internal fluctuation B.

This gives three useful conclusions.

1. Such a coefficient can be factored out of the actual local R fiber before its three localization joins. Its finite source polynomial, prescribed boundary data, saved incoming factors, and original mesh are unchanged. The statement applies with the actual selectors and denominators; it does not require a product selector support.
2. Every internal derivative of its pullback along this native fiber vanishes. When the same retained rows belong to the output record, its complete conditional covariance packet, including an existing pushforward boundary defect, also vanishes. This is an exact cancellation for this sector, not an estimate by a small outer density.
3. The native joins themselves do not supply this conclusion for an arbitrary previously generated marked coefficient. R-II's joining operation is the operation in III (2.14), not a literal union of row sets. The distinction becomes material at the three boundary layers in (1.61)–(1.63).

There is also a quantitative internal-direction bound for the actual nonzero-fluctuation **reference** fiber on a complete protected top-scale record. It follows from the 1985 Proposition 9 and extends the restricted SG11 estimate beyond its $B = 0$ endpoint. It is not a bound for the numerator's three-join field.

These results do not close AE8/LM47 for all actual mixed histories. They identify a sector in which admission can be done exactly, and a different native fiber on which an internal derivative estimate is available. They do not prove the required common complex chart for an unprotected inherited marked readout.

154.3 The unchanged raw-row algebra in the actual numerator

Fix an original history and one of its active local R operations. In the notation of R-II (1.50), let $U'_k(B,\xi)$ be its actual numerator background; ξ includes the retained fields. Let $U_k(\xi)$ be the new background used in (1.58). Define $\mathcal{C}$ to be the **literal common retained rows outside** Λ, with the following convention.

A row consists of its raw averaging prefix, target value, endpoint frames, any junction/site constraints, and every prescribed exterior value read by that prefix. Two rows are declared common only in this recorded sense. They are not identified merely because their projected bond labels coincide. The necessary compensating gauge in R-II (1.61) is retained when making this comparison.

Write $M_{\mathcal{C}}$ for these raw constraints in their common recorded frame, and $v_{\mathcal{C}} = V''|_{\mathcal{C}}$ for the retained target array. R-II (1.50), together with the sentence following it, gives

$$M_{\mathcal{C}}(U'_k(B,\xi)) = v_{\mathcal{C}}(\xi) = M_{\mathcal{C}}(U_k(\xi)). \tag{PE12.1}$$

This is a constraint identity for the actual numerator; B has not been set to zero. The exterior values and frames on both sides are the ones already present in the original operation.

For a function F of precisely this completed target record, define

$$\widehat{F}(U) = F(M_{\mathcal{C}}(U)). \tag{PE12.2}$$

Then

$$\widehat{F}(U'_k(B,\xi)) = F(v_{\mathcal{C}}(\xi)) = \widehat{F}(U_k(\xi)). \tag{PE12.3}$$

This proof uses the actual constraint equations once; it does not use a weak Sobolev estimate, a minimum-energy inequality, or an assertion that all fields appearing in a localization formula coincide.

154.3.1 A concrete original-mesh example

Suppose a finite fine plaquette word, and every bond needed for its recorded current and boundary partners, consists of level-zero rows in $\mathcal{C}$. Level-zero rows are identity constraints. The word is therefore exactly the same word in the retained fine bonds V''. For example, with its original base point and orientation,

$$P_p(U'_k) = V''_{b_1} V''_{b_2} (V''_{b_3})^{-1} (V''_{b_4})^{-1}. \tag{PE12.4}$$

The same observation applies to a finite collection of such words and their complete current stars. It fails as soon as a required star or prefix uses a row which is not common and retained. In particular, it is not enough that the central plaquette or the visible carrier lies outside Λ.

If an already generated coefficient has explicitly been stored as a function of this same completed raw target record, PE12.3 applies to that coefficient too. No claim is made that every coefficient generated by the marked induction has this factorization. Establishing it for a particular coefficient means checking its complete saved dependence record, not just its nominal support.

154.3.2 Internal derivatives and source labels

On any analytic branch of this native constrained fiber, its defining constraint equations remain PE12.1. Thus, for every nonempty multi-index α in the actual internal B coordinates,

$$\partial_B^{\alpha} \widehat{F}(U'_k(B,\xi)) = 0. \tag{PE12.5}$$

The assertion is simultaneous in all these internal directions. There is no sum over coordinate directions to estimate, and no hidden dependence on the fine coordinate count. Outer derivatives of $F(v_{\mathcal{C}}(\xi))$ need not vanish.

Let η_I denote the original finite, commuting nilpotent source words. Every identity above holds coefficientwise for

$$F(\eta, v_{\mathcal{C}}) = \sum_I \eta_I F_I(v_{\mathcal{C}}). \tag{PE12.6}$$

It therefore preserves the source word, original offsets, component labels, and saved prefactor of each coefficient. It neither supplies an absent factor of the fine spacing nor changes a physical current exponential to its Ward completion.

154.4 Factoring before the original localization joins

Keep the actual selector record, retained parameters, and normalization. For any one of the local native expressions write, schematically but without changing its variables,

$$\mathcal{T}_\omega[H](z) = D_\omega(z)^{-1} \int K_\omega(z, d\xi)\, H(z,\xi). \tag{PE12.7}$$

Here K_ω includes its original constraint and gauge factors, Haar/coarea factors, inherited effective density, and selectors. The original D_ω is used; no scalar normalization is set to one. The formula is an algebraic shorthand for the actual finite fiber operation, not an assertion that the selector support is a Cartesian product. On the left of the local operation the completed array $v_{\mathcal{C}}$ is retained.

For every integrand H, PE12.3 implies the exact module identity

$$\mathcal{T}_\omega[\widehat{F}(U'_k)H] = F(v_{\mathcal{C}})\mathcal{T}_\omega[H]. \tag{PE12.8}$$

It holds first for the raw numerator. Division by the same genuine nonzero denominator gives the normalized identity. If a denominator has source dependence, the identity still holds in the finite source algebra with that dependence retained; any corresponding source factor in another numerator or denominator must be kept explicitly. PE12.8 does not authorize dropping an absolute scalar increment.

The useful ordering is to use PE12.8 **before** localizing the kernel. In R-II (1.61)–(1.64), the localization parameters act on the extension functions H'_k, H_{B_1}, H_k, while the outer retained target parameters remain parameters. After the exact factorization, the original localization can be performed on the same kernel with $F(v_{\mathcal{C}})$ left outside. On any common complex domain on which the retained expression is holomorphic, its extension is simply

$$F_{\mathrm{prot}}(z, \mathbf{s}) = F(v_{\mathcal{C}}(z)). \tag{PE12.9}$$

It is independent of all three sets of localization walls simultaneously. All mixed derivatives having at least one such wall derivative vanish. The complete finite source expansion is unchanged, since PE12.8 is a coefficientwise equality before the expansion.

There is an important distinction for current partners. An outer current direction can vary $v_{\mathcal{C}}$, the kernel, and its denominator. PE12.8 does not commute that direction through $\mathcal{T}_\omega$ without the complete product and quotient responses. The transport cost one below is the cost for admission of the function-valued coefficient and for its internal B jets. It is not a bound for derivatives of the native kernel with respect to retained variables. Such derivatives keep the effective score, live reference response and normalization commutator exactly as in DT12.

This is an exact source-preserving representation of the protected sector using the same integration maps. It is **not** the assertion

$$M_{\mathcal{C}}(\Gamma^R_{\omega,\mathbf{s}}) = v_{\mathcal{C}} \tag{PE12.10}$$

for every individual off-shell localized preparation. PE12.10 would require an additional check of the localized nonlinear constraints. It does not follow just from equality at the physical wall vertex. The factorization removes the need to assert PE12.10 for the protected sector; it does not prove PE12.10 for other coefficients.

154.4.1 Quantitative grades and their domain limitation

Suppose the existing coefficient bound is

$$|F_I(v)| \le \Gamma_I A_I \qquad (v \in \mathcal{V}_I), \tag{PE12.11}$$

where Γ_I is its saved scale gain and $\mathcal{V}_I$ is its actual previously proved target domain. Whenever the retained projection of the common complex domain lies in $\mathcal{V}_I$, PE12.9 gives the same bound with exactly Γ_I. Its transport cost is one. There is no new scale label, reserve charge, or component count. This does not prove the domain inclusion if an old coefficient was known only on the auxiliary M8 weak chart.

For the explicit finite-word sector, no branch of a weak potential chart is needed. If

$$F_I(v) = \Gamma_I a^{-d_I} \sum_{\nu} c_{I,\nu} W_{I,\nu}(v), \qquad \ell_I = \max_{\nu} \operatorname{length}(W_{I,\nu}), \tag{PE12.12}$$

and the actual common domain has $\|v_b\|, \|v_b^{-1}\| \le K$ on every recorded row, then

$$|F_{I,\mathrm{prot}}| \le |\Gamma_I| a^{-d_I} \Big(\sum_{\nu} |c_{I,\nu}| \Big) K^{\ell_I}. \tag{PE12.13}$$

For traces include the fixed matrix dimension in the displayed word coefficient norm. This is a bound on the simultaneous wall domain, because the words themselves are wall-independent. It uses the actual row bound K, not compactness for a complex field. The physical power a^{-d_I}, the word lengths, and the coefficient sum all remain explicit. Thus it supplies no missing original-mesh birth gain.

For general inherited analytic coefficients, PE12.11 still needs its actual domain inclusion. The algebraic result should not be reported as a proof of that analytic premise.

154.6 Why the three actual joins do not prove general admission

The operation denoted by the joining symbol in R-II is defined in III (2.14). The new determining set agrees with the old one on a protected interior, and adds/replaces the appropriate rows in a boundary region. It is not the set-theoretic union of all old and all new rows.

This distinction is explicit at every stage of the actual R-II construction.

Native stage	Protected coincidence and original boundary data
(1.61), printed p.373	Y_1^0 is where $B_k'(Y_1)$ and B_k' coincide. The complement is a boundary layer of width at most $2M$ at its own scale. The compensating gauge u_k' must be retained.
(1.62), printed pp.373–374	Y_2^0 is where $B_1(Y_2)$ and B_1 coincide. The second localized extension depends on the next background and its complete boundary data.
(1.63)–(1.64), printed p.374	The final join is $B_k(Y_3)$. Its exterior target and relaxed interior target are different recorded pieces; the localized H_k is then composed into the preceding joins.

Thus a row-factor argument made within an individual joined expression must check the whole row record against each relevant coincidence region, including its boundary frame. Its old raw readout does not follow from a statement about the visible carrier alone. Moreover, the physical identities of the constrained minima do not by themselves say that arbitrary independent localized extension functions preserve those constraints.

R-II (1.65), p.375, states an analytic inclusion for the prepared composite into the final R-I range. It explicitly uses the final preparation index. The following paragraph verifies inclusion in the domains of the old **native effective-action terms** by their inductive hypotheses. It does not state that an arbitrary old marked coefficient constructed only on M8's unweighted weak chart belongs to that class. In particular it does not give the weighted AE8 curvature sum for the actual complete marked readout.

The protected factorization of Sections 3–5 is therefore a way of completing those transitions whose coefficient really belongs to the unchanged-row algebra. It is not a replacement for the missing admission estimate at a transition which touches the source's full record.

154.8 An original boundary projection that must be retained

R-II (1.54)–(1.55), printed pp.370–371, contains an actual projection correction for the current of the joined background $U_{1,2}$. This is useful evidence about which cancellations are native, and which cannot be assumed for arbitrary marked currents.

The native propagator G_k'' annihilates the adjoint averaging rows, and the current J_1 satisfies constrained criticality in the interior. Cutting it off without a correction fails at the original boundary layer. The source inserts the projection built from the actual averaging map and its right inverse, then obtains (1.55): a term supported where the cutoff varies, together with the explicitly retained boundary current $\theta J_{1,2}$. The cutoffs and boundary layer have their original scale-dependent width; the layer is not removed from the record.

The mechanism has a useful exact finite-matrix form. If Q is the actual linearized raw averaging map, $GQ^* = 0$, and χ_b, χ_c are the chosen fine-bond and row cutoffs, then

$$(\chi_b Q^* - Q^*\chi_c)_{b,c} = (\chi_b - \chi_c)(Q^*)_{b,c}. \tag{PE12.25}$$

Only original prefixes crossing the cutoff transition contribute. Using orthonormal coordinates for the original weighted scalar products, its operator norm is bounded by

$$\|\chi_b Q^* - Q^*\chi_c\|_{2\to 2} \le \left[\sup_b \sum_c |\chi_b - \chi_c| \|(Q^*)_{b,c}\| \ \sup_c \sum_b |\chi_b - \chi_c| \|(Q^*)_{b,c}\|\right]^{1/2}. \tag{PE12.26}$$

This is the weighted Schur estimate applied to the actual recorded matrix. The weights, row lengths, number of rows, and number of bond components are all included in that matrix and in the two displayed sums. If a Lipschitz cutoff is used, one may bound $|\chi_b - \chi_c|$ by its actual Lipschitz constant times the corresponding prefix diameter. No uniform bound on those sums is asserted here without the original row geometry.

This shows precisely why the projection in native (1.55) matters: it converts constrained criticality into a boundary-supported commutator, not into an assertion that the unprojected current vanishes. The current J_1 in that argument is the current of a specified constrained background. A physically prescribed inherited degree-two source is not identified with that current merely because both obey some Ward identities. Thus (1.55) is not an automatic cancellation of R12.

155.1 What is fixed by the original prescription

This note continues WC11.45–46 using the physical sources that were actually fixed in MT1 and AI35d. It does not replace them by a Ward-flow source. The bare sources are

$$O_{H,a}(f) = g^{-2} H_a(f), \qquad O_{C,a}(f) = g^{-3} C_a^\sharp(f), \qquad \beta = c_\beta g^{-2}. \tag{PD12.1}$$

Here the globally smooth representatives, exact pointed hypercubic averages, shifted tests, compact fluxes, and cubic completion are exactly those in AI and MT. In particular, $C_a^\sharp$ includes the complete lattice remainder in AI32a–b. It is not replaced by an onsite continuum cubic. PA4 and MT21 change an analytic extension in verified primitive slots; their all-one physical endpoint is still the same observable in PD12.1.

XP3 fixes the original finite-source multiplier for two labels as

$$M(\eta, U) = \exp_{\mathcal{J}}\{\eta_1 O_1(U) + \eta_2 O_2(U)\}, \qquad \eta_1^2 = \eta_2^2 = 0. \tag{PD12.2}$$

Consequently the **bare physical action coefficient is**

$$A_{12}^{\mathrm{bare}} = 0, \qquad [\eta_1 \eta_2] M = O_1 O_2. \tag{PD12.3}$$

There is no mixed-source compensator in PD12.2 unless one was separately included in the physical prescription. A field-independent subtraction from a single O_i changes its mean but does not supply a mixed-source counterterm. Subsequent exact integration does generate a nonzero action coefficient. It is determined by the joint physical product, not chosen to make an isolated EOM current disappear.

The all-history EOM projection is not specified uniquely by MT1. A decomposition of a scalar function into a current and an ordinary part is not unique. Sections 2–4 use explicit identities already present in AI, so their current choices and their compensating ordinary pieces are completely specified. They compute a correlated packet for the actual source in PD12.2. They do not claim that an undocumented projection at every inherited transition has already been identified with that choice.

The new quantitative results are as follows.

- The actual H source admits an explicit two-source current/ordinary compensation coefficient, including the Casimir companion. Its real compact bound spends the same single plaquette-energy reserve and retains the physical factor a^{-4}.
- For the original Wilson measure, two copies of the actual $g^{-2}H_a$ source have a complete coincident connected coefficient at least cT^4a^{-8}. This is a statement about the original H observable, not about an independently exponentiated current.
- The absolute scalar mass of the explicitly compensated bare H contact is bounded by $Ca^{-4}N_Y[1+\sqrt{\log(2+\beta)}]$, with its actual overlap count N_Y. A matching upper bound for the full physical H variance still requires control of its ordinary connected sector. The note states that missing estimate explicitly.

None of these conclusions proves all-history MT36, AE8/LM47, or complete S94.

155.2 Exact decompositions of the two physical marks

Use the original finite Wilson density

$$\rho(U) = e^{-\beta S_0(U)}, \qquad d\mu_\beta = Z_\beta^{-1}\rho\, d\lambda, \qquad S_0 = \sum_p s_p, \qquad s_p = N - \Re\,\mathrm{tr}\, U_p. \tag{PD12.4}$$

The invariant derivatives X_i, physical force $J_i = a^{-3}X_i S_0$, and $\kappa = (\beta a^3)^{-1}$ are AI3. For a Lie-algebra bond coefficient q, define

$$\mathscr{E}_a[q] = a^4 \sum_i \{J_i q_i - \kappa X_i q_i\}, \qquad V[q] = \kappa a^4 \sum_i q_i X_i. \tag{PD12.5}$$

Thus $D_{V[q]}\rho = \rho\mathscr{E}_a[q]$, with $D_V u = -\operatorname{div}_\lambda(uV)$. Coupling normalizations may be included in q. All derivatives of q then remain in PD12.5.

155.2.1 The H mark, including its scalar companion

For one literal pointed term of AI34, set

$$\begin{aligned} D_a(f) &= a^4 \sum_{b,A} f_b \sum_{\mu\neq\nu_b} (J^A_{b;\mu})^2, \\ P_a(f) &= a^4 \sum_{b,A} f_b (J^A_b)^2, \\ B_a(f) &= \kappa a^4 \sum_{b,A} f_b X^A_b J^A_b, \qquad H_a(f) = D_a(f) - \tfrac{1}{3}P_a(f). \end{aligned} \tag{PD12.6}$$

The exact AI34 decomposition of the physical H source is

$$\begin{aligned} q_{H,f} &= -\tfrac{1}{3}g^{-2} f J, \\ t_{H,f} &= \mathscr{E}_a[q_{H,f}] = -\tfrac{1}{3}g^{-2}\{P_a(f) - B_a(f)\}, \\ r_{H,f} &= g^{-2}\{D_a(f) - \tfrac{1}{3}B_a(f)\}, \\ O_{H,a}(f) &= t_{H,f} + r_{H,f}. \end{aligned} \tag{PD12.7}$$

Both appearances of the lower scalar $B_a(f)$ are required. Their opposite contributions cancel in the sum, giving the original H exactly. The Casimir formula fixes the scalar completely:

$$B_a(f) = C_F \beta^{-1} a^{-2} \sum_p f_p (N - s_p), \qquad f_p = \sum_{b\in\partial p} f_b. \tag{PD12.8}$$

The constant term in PD12.8 is retained. Each of the four original plaquette incidences is present; the test is not replaced by its value at a common center.

For the fixed pointed hypercubic completion, apply PD12.6–8 to every member of the original finite average, with its transformed bond, test, and transport record, and then take that average. The vector fields and ordinary parts are averaged together. At degree two the bilinear formulas below contain the double average. They do not identify different pointed entries before their test offsets have been kept. For a constant test the global $H_a(1)$ is already invariant under this average.

155.2.2 The C-sharp mark, including its descendant and remainder

Write V_f^{sp} for the **spacetime** test-gradient coefficient in AI17,

$$(V_f^{\mathrm{sp}})_\nu(x) = \sum_\mu (D_\mu^* f)(x) T_\mu^{-1} X_{\mu\nu}(x). \tag{PD12.9}$$

This notation distinguishes it from the configuration-space vector field $V[q]$. The exact identity AI32d gives

$$\begin{aligned} q_{C,f} &= g^{-3}(fJ + V_f^{\mathrm{sp}}), \\ t_{C,f} &= \mathscr{E}_a[q_{C,f}], \\ r_{C,f} &= g^{-3}\left\{-\tfrac{1}{2}A_a^{\mathrm{sym}}(f) + \tfrac{1}{2}\mathcal{C}_a(f) + \tfrac{1}{4}Q_a(\Delta_a f)\right\}, \\ O_{C,a}(f) &= t_{C,f} + r_{C,f}. \end{aligned} \tag{PD12.10}$$

Here $\mathcal{C}_a(f) = 2\kappa a^4 \sum_i X_i(fJ_i + (V_f^{\mathrm{sp}})_i)$ is the literal AI29 counterterm. Its identity and Wilson-action terms are both retained. The last line in PD12.10 follows by solving AI32d for $C_a^\sharp$; its coefficient of $Q_a(\Delta_a f)$ is $1/4$, not $1/2$. No $\mathcal{R}_a$ is lost: that remainder was included in the definition of $C_a^\sharp$. If one instead chooses the distinct bare observable C_a^{split}, the additional $g^{-3}\mathcal{R}_a/2$ belongs in its ordinary part.

Every test derivative and shifted curvature in PD12.9–10 remains in its original physical stencil. The whole identity is averaged for the pointed completion, as in AI4.1. In particular the ordinary part is not simply a lower-dimensional constant. It contains the derivative scalar and the quadratic descendant which compensate the low field jets of its current part.

155.4 Explicit H–H coefficient with all lower responses

The H pair permits a more concrete formula. Use dimensionless forces and sums

$$\begin{aligned} F_i &= X_i S_0, \qquad F_{i;\mu} = X_i S_0^{(\mu\nu_i)}, \qquad F_i = \sum_{\mu\neq\nu_i} F_{i;\mu}, \\ P[f] &= \sum_i f_i F_i^2, \qquad D[f] = \sum_i f_i \sum_{\mu\neq\nu_i} F_{i;\mu}^2, \\ \ell[f] &= \sum_i f_i X_i^2 S_0, \qquad W_f = \sum_i f_i F_i X_i, \qquad \sigma = g^{-2}a^{-2}. \end{aligned} \tag{PD12.20}$$

The f_i are color independent and field independent. All color sums are literal. The physical quantities in Section 2 are

$$\begin{aligned} O_H(f) &= \sigma\{D[f] - P[f]/3\}, \\ t_H(f) &= -\tfrac{\sigma}{3}\{P[f] - \beta^{-1}\ell[f]\}, \\ r_H(f) &= \sigma\{D[f] - (3\beta)^{-1}\ell[f]\}, \\ V_H(f) &= -\frac{\sigma}{3\beta} W_f. \end{aligned} \tag{PD12.21}$$

Substitution in PD12.13, with its original bare $A_{12} = 0$, yields

$$\boxed{\begin{aligned} b_{HH}(f,h) = \frac{\sigma^2}{9\beta}\Big\{ &\tfrac{1}{2}(W_f P[h] + W_h P[f]) \\ &- 3(W_f D[h] + W_h D[f]) \\ &+ \frac{1}{2\beta}(W_f \ell[h] + W_h \ell[f])\Big\}. \end{aligned}} \tag{PD12.22}$$

In particular, the sign of the scalar-companion response in the last line is positive. The same response has the opposite sign in the isolated Ward defect

$$R_{HH}(f,h)=\frac{\sigma^2}{18\beta}\left\{W_fP[h]+W_hP[f]-\beta^{-1}(W_f\ell[h]+W_h\ell[f])\right\}. \tag{PD12.23}$$

For equal uniform tests, let $W=W_1$, $P=P[1]$, $D=D[1]$, and $\ell=\ell[1]$. The exact Casimir identity is

$$\ell=4C_F(N\#p-S_0),\qquad W\ell=-4C_FP. \tag{PD12.24}$$

Consequently

$$b_{HH}(1,1)=\frac{\sigma^2}{9\beta}\{WP-6WD-4C_F\beta^{-1}P\}. \tag{PD12.25}$$

This is an exact nonlinear Wilson coefficient. The WD term is the response of the actual ordinary directional-square part. Omitting it reproduces a separately sourced-current test, not the physical H packet.

155.4.1 A real compact, one-reserve bound for the actual coefficient

Let $Y=Y(f,h)$ be the literal union of the plaquette stars and shifted pointed stencils occurring in nonzero terms of PD12.22. A term in $W_fD[h]$, for example, requires a differentiated f-bond and an h-bond in a common plaquette. If there are no such contacts in either order, PD12.22 vanishes. Let

$$E_Y=\beta\sum_{p\in Y}s_p. \tag{PD12.26}$$

Then, for real compact fields,

$$\boxed{|b_{HH}(f,h)|\le C_Nc_\beta^{-2}a^{-4}(1+\beta^{-1})\|f\|_\infty\|h\|_\infty E_Y.} \tag{PD12.27}$$

Proof. A plaquette first derivative satisfies $|X_is_p|\le C_N\sqrt{s_p}$ on its actual star, and every fixed second derivative of a plaquette action is bounded on the compact group. A force or directional force therefore obeys

$$|F_i|^2+\sum_\mu|F_{i;\mu}|^2\le C_N\sum_{p\ni b_i}s_p. \tag{PD12.28}$$

In $W_fP[h]$ and $W_fD[h]$, the derivative of a force is zero unless the two links meet a common plaquette, and is bounded when it is nonzero. Apply $2|uv|\le|u|^2+|v|^2$ to the remaining two force factors. The number of neighboring plaquettes, links, plane slots, and colors is fixed. Thus each of these two expressions is bounded in absolute value by $C_N\|f\|_\infty\|h\|_\infty\sum_{p\in Y}s_p$.

The scalar response is not estimated as an unspecified third derivative. Its actual Casimir formula gives

$$\ell[h]=C_F\sum_p h_p(N-s_p),\qquad W_f\ell[h]=-C_F\sum_i f_iF_i\sum_{p\ni b_i}h_pX_is_p. \tag{PD12.29}$$

Since $|h_p| \le 4\|h\|_\infty$, the same bound follows from PD12.28 and the first plaquette-derivative bound. Combining the six terms in PD12.22 gives $C\sigma^2\beta^{-1}(1+\beta^{-1})\|f\|_\infty\|h\|_\infty \sum_Y s_p$. Use $\sigma^2\beta^{-2} = c_\beta^{-2}a^{-4}$ and PD12.26. □

No new energy has been introduced. If this estimate is combined with a selected Gaussian or cold estimate, its polynomial factor is paid from that estimate's already retained common reserve. Neither PD12.27 nor a finite source degree creates a new reserve. The bound does not hold on arbitrary complex wall arguments by positivity; those arguments require their actual common domain and a separate complex estimate.

155.4.2 An absolute scalar bound with the literal contact count

Let $N_Y = \#\{p \in Y\}$, counting the original plaquettes of the completed overlap, rather than the volume of an unrelated coarse box. On an isotropic periodic torus of side T, for all sufficiently large β,

$$\boxed{\mathbb{E}_{\mu_\beta}|b_{HH}(f,h)| \le C_{N,c_\beta}\, a^{-4} N_Y \|f\|_\infty \|h\|_\infty [1 + \sqrt{\log(2+\beta)}].} \tag{PD12.30}$$

The proof uses two elementary inputs from the same measure:

$$\beta \mathbb{E} F_i^2 = \mathbb{E} X_i^2 S_0 \le C_N, \qquad \mathbb{E} s_p \le C_N \frac{\log(2+\beta)}{\beta}. \tag{PD12.31}$$

The first is exact Haar integration by parts. The second is the partition-function and symmetry estimate proved again in Section 6.1 below. Equation PD12.28 gives $\mathbb{E}F_{i;\mu}^2 \le C_N \log(2+\beta)/\beta$. Hence each nonzero term in $W_f P[h]$ has absolute expectation at most $C\|f\|\|h\|/\beta$, whereas each in $W_f D[h]$ is at most $C\|f\|\|h\|\sqrt{\log(2+\beta)}/\beta$. This follows by Cauchy–Schwarz for its two force factors and the compact bound for its intervening second derivative. The finite incidence count sums those terms to CN_Y. PD12.29 gives at most the latter bound for $W_f \ell[h]$, followed by its additional β^{-1} in PD12.22. Finally $\sigma^2\beta^{-2} = c_\beta^{-2}a^{-4}$, proving PD12.30.

Writing $|Y|_a = a^4 N_Y$ makes its physical-volume form

$$\mathbb{E}|b_{HH}(f,h)| \le C a^{-8} |Y|_a \|f\|_\infty \|h\|_\infty [1 + \sqrt{\log(2+\beta)}]. \tag{PD12.32}$$

This is an actual absolute-mass estimate for the compensated bare contact. It is not yet an all-history estimate for the effective ordinary covariance, effective scores, or derivatives of a previously generated coefficient.

155.5 Complete kernel responses and actual inherited coefficients

155.5.1 What is differentiated

Let $\mathsf{K}_U(d\zeta)$ denote the complete original auxiliary construction as in WC11.11–17: the selector alternatives are all present; each matched native denominator remains in its original numerator slot; all old variables are retained until their prescribed integration. The complete kernel has $\mathsf{K}_U 1 = 1$. An individual selected term need not have unit mass.

For a carrying function $\Psi(U,\zeta)$, write $\Phi(U) = \mathsf{K}_U \Psi(U,\cdot)$, interpreted weakly when constraints or selectors are distributional. The exact degree-two identity for the original source is

$$\int \rho\, O_1 O_2\, \Phi\, d\lambda = \int \rho\big\{(r_1 r_2 + b_{12})\Phi + r_1 V_2\Phi + r_2 V_1\Phi + \tfrac{1}{2}(V_1V_2 + V_2V_1)\Phi\big\}\, d\lambda. \tag{PD12.33}$$

For the bare source, b_{12} is PD12.13 with $A_{12} = 0$. If an actual inherited A_{12} is already present, its ordinary multiplicative contribution is added before the same calculation. No derivative acts only on the displayed classical action while ignoring the rest of the carrying function.

If a degree-two current word is extracted as in PD12.13a, its contribution to the right side is $-t_{12}\Phi + L_{12}\Phi$ before integration, with $L_{12} = V_{12}$ in the first-order case. These cancel by adjointness only as the complete packet. The ordered second-current term and every lower source word retain their original coefficients.

The full weak product rule is

$$V_i(\mathsf{K}\Psi) = (V_i\mathsf{K})\Psi + \mathsf{K}(V_i\Psi), \tag{PD12.34}$$

and, for $S_{12} = \frac{1}{2}(V_1V_2 + V_2V_1)$,

$$S_{12}(\mathsf{K}\Psi) = (S_{12}\mathsf{K})\Psi + (V_1\mathsf{K})(V_2\Psi) + (V_2\mathsf{K})(V_1\Psi) + \mathsf{K}(S_{12}\Psi). \tag{PD12.35}$$

The vector fields can have variable coefficients; their ordered compositions retain those derivatives. Equations PD12.34–35 include every original constraint response, cutoff-boundary distribution, coarea derivative, reference derivative, normalization commutator, and inherited effective score. On a verified matched fiber the quotient is differentiated as the original N/D; its denominator is not frozen a second time or replaced by a different conditional mass.

If $\Psi = F(U)$ is the actual old physical carrying function, the complete auxiliary integral gives $\mathsf{K}F = F$, and PD12.33 reduces to the bare Haar identity. For $F = 1$, all complete kernel responses vanish and one obtains PD12.16. This removes auxiliary Ward responses, not the physical coefficient $\mathrm{Cov}(O_1, O_2)$. If the output is retained or constrained, the weak derivatives in PD12.33 remain. One cannot set them to zero using only normalization of an unconstrained closed kernel.

155.5.2 The actual mixed action after a fixed source-free integration

Let $\mathcal{T}$ be a finite composition of the original, source-free maps, with all old mark slots retained. Its unnormalized source coefficients for PD12.2 are

$$N_0 = \mathcal{T}(\rho), \qquad N_i = \mathcal{T}(\rho O_i), \qquad N_{12} = \mathcal{T}(\rho O_1 O_2). \tag{PD12.36}$$

On an actual clean output where $N_0 \neq 0$, its physical source action satisfies

$$A_i^{\mathrm{out}} = \frac{N_i}{N_0}, \qquad \boxed{A_{12}^{\mathrm{out}} = \frac{N_{12}}{N_0} - \frac{N_1 N_2}{N_0^2}.} \tag{PD12.37}$$

These are the actual coefficients which must enter WC11.46 at that output. XP18 and XP40 give the same expression after any exact deterministic-tag split. In particular all old tagged A_{12} terms,

jointly integrated proper differences, and their covariance are included. There is no freedom to insert a compensator in PD12.37 without changing the source functional.

When the map is a positive disintegration, PD12.37 is the genuine conditional covariance of the original physical observables. In a still-live selected numerator, retain PD12.36 and the joint product; a positive normalized conditional measure or an inverse N_0 is not asserted there. In a complex or signed clean representation, PD12.37 remains an algebraic ratio identity where its original nonzero body has been established, but positivity conclusions do not follow from that identity.

At the terminal physical integral, with $O_1 = O_2 = O_H(f)$,

$$[\eta_1\eta_2] \log \frac{Z(\eta)}{Z_\beta} = \mathrm{Var}_{\mu_\beta}(O_H(f)). \tag{PD12.38}$$

This expression is independent of the chosen current decomposition and of any exact source-preserving representation. It supplies a direct test of whether a proposed total scalar cancellation is compatible with the original prescription.

155.6 A nonlinear lower bound for the actual H source

Every ordered variance inequality in this section and Section 7 uses real-valued test functions. The earlier absolute-value contact estimates also allow complex test weights.

Fix $G = SU(N)$ with $N \geq 2$, and $c_\beta > 0$. Let the lattice be an isotropic four-torus with $n = T/a$ sites per direction. Assume n exceeds a fixed geometric constant so the finite neighborhoods used below embed without wrapping. Use the original, symmetrized constant-test observable in MT1.

Define

$$G_H(U) = \sum_{b,A} \left\{ \sum_{\mu \neq \nu_b} (F^A_{b;\mu})^2 - \tfrac{1}{3}(F^A_b)^2 \right\}, \qquad O_H(1) = g^{-2}a^{-2}G_H. \tag{PD12.39}$$

The summands are exactly $\sum_\mu |F_{b;\mu} - F_b/3|^2$, so $G_H \geq 0$. This is the full physical H insertion, with its original force definition and coupling normalization.

Theorem. There are $\beta_0, c > 0$, depending only on the fixed group, normalization, and local lattice geometry, such that

$$\boxed{\mathrm{Var}_{\mu_\beta}(O_H(1)) \geq cT^4a^{-8}} \qquad (\beta \geq \beta_0). \tag{PD12.40}$$

This lower bound uses the exact nonlinear Wilson law and is uniform in n and $\beta \geq \beta_0$. It is a coincident two-source assertion. It does not say that separated physical correlation functions fail to have a renormalized limit.

155.6.1 Uniform probability of a locally near-flat bath

First,

$$-\log Z_\beta \leq C\,\#E\,\log(2+\beta). \tag{PD12.41}$$

Restrict all links to Haar balls of radius $c\beta^{-1/2}$. Each plaquette action is then at most $C\beta^{-1}$, and each link-ball volume is at least $c'\beta^{-\dim G/2}$. Their product supplies PD12.41. Since $\mathbb{E}_\beta S_0$ decreases with β,

$$\frac{\beta}{2}\mathbb{E}_\beta S_0 \le \log Z_{\beta/2} - \log Z_\beta \le -\log Z_\beta. \tag{PD12.42}$$

Translation and coordinate permutation invariance of the isotropic torus yield the second inequality in PD12.31, uniformly for each plaquette. For any fixed contractible plaquette patch Q, Markov's inequality gives

$$\mu_\beta\{s_p \le \varepsilon^2 \text{ for all } p \in Q\} \ge 1 - C_Q\varepsilon^{-2}\frac{\log(2+\beta)}{\beta}. \tag{PD12.43}$$

Only a fixed microscopic patch is involved.

155.6.2 Independent free links and additivity of the actual H observable

Join two lattice links in the interaction graph if they lie in a common plaquette. Choose a family $\mathcal{F}$ of links mutually separated by graph distance at least five. Since this graph has fixed bounded degree, a greedy packing gives

$$|\mathcal{F}| \ge c_0 n^4. \tag{PD12.44}$$

Condition on every link outside $\mathcal{F}$, denoting the bath by $\mathfrak{b}$. No plaquette contains two free links, so the conditional Wilson law factors as the product of its one-link laws. Every summand of G_H is a quadratic expression in forces centered on one link and reads only its plaquette star. Its determining set therefore meets at most one free link. Thus, with the bath fixed,

$$G_H = G_0(\mathfrak{b}) + \sum_{b\in\mathcal{F}} G_{H,b}(U_b;\mathfrak{b}), \tag{PD12.45}$$

and conditional independence implies

$$\operatorname{Var}_{\mu_\beta}(G_H) \ge \sum_{b\in\mathcal{F}} \mathbb{E}_{\mu_\beta}\operatorname{Var}_{b,\beta}(G_{H,b} \mid \mathfrak{b}). \tag{PD12.46}$$

All terms of the full H observable are present in this decomposition. It is not a conditional variance calculation for one current term isolated from its ordinary companions.

Call a bath good near b if there exists a unitary vertex gauge putting all its links in the finite determining neighborhood of $G_{H,b}$ within a sufficiently small fixed δ of the identity. This property depends only on the conditioned links. A fixed contractible cube containing that neighborhood has a spanning-tree gauge. If its plaquettes all satisfy $s_p \le \varepsilon^2$, every bounded-length tree-cycle holonomy is within $C_Q\varepsilon$ of the identity, by a product of unitary plaquette holonomies and telescoping. Taking $C_Q\varepsilon < \delta$, PD12.43 gives

$$\mu_\beta(\mathfrak{b} \text{ good near } b) \ge \tfrac{1}{2} \qquad (\beta \ge \beta_0(\delta)). \tag{PD12.47}$$

The gauge witnessing this implication may be found using a full configuration. The event itself asserts existence of a gauge for the bath links and is bath measurable: it can be defined by minimizing their

maximum transformed distance over the compact finite vertex gauge group. No free-link value is frozen in the event.

155.6.3 The exact one-link H calculation

Put every bath link in that determining neighborhood equal to the identity and vary the single free link $U_b = U$. Put $s(U) = N - \Re \operatorname{tr} U$. There are six incident plaquettes.

At the free bond, each of its three plane forces is $2\nabla s(U)$, whereas its total force is $6\nabla s(U)$. Its H summand is therefore

$$3 \cdot 4|\nabla s(U)|^2 - \tfrac{1}{3} \cdot 36|\nabla s(U)|^2 = 0. \tag{PD12.48}$$

The other three bonds in each incident plaquette are distinct from the corresponding bonds in the other five plaquettes, provided n exceeds the fixed local size. They give eighteen other bonds. Each has one nonzero plane force, of norm $|\nabla s(U)|$, and the same total-force norm; the other plane forces vanish. Each therefore contributes $(1 - 1/3)|\nabla s(U)|^2 = 2|\nabla s(U)|^2/3$.

All unchanged plaquettes have zero first force. Unit adjoint transports and possible orientation signs preserve the squared norm. Consequently the exact variable part is

$$\boxed{G_{H,b}^{\mathrm{flat}}(U) = 12|\nabla s(U)|^2 + \text{a bath constant.}} \tag{PD12.49}$$

The coefficient is $18 \cdot 2/3 = 12$, not the coefficient 54 of the different total-current-square observable in WC11.38. This is where the actual physical source changes the calculation.

In exponential coordinates x at the identity, $s(e^x) = \frac{1}{2}\langle x, H_s x\rangle + O(|x|^3)$, where H_s is the positive definite matrix fixed by the generator normalization. Thus the Hessian of $12|\nabla s|^2$ is strictly positive at the identity. On a sufficiently small fixed chart it is bounded below by cI. The function $G_{H,b}$ is a finite expression in a fixed number of plaquette words. Continuity therefore preserves this Hessian lower bound throughout a smaller chart for every sufficiently small good bath, after its unitary gauge. Constants are independent of the torus and β.

155.6.4 Conditional one-link variance

On a good bath the conditional one-link Wilson action has the exact form

$$S_b(U; \mathfrak{b}) = 6N - \Re \operatorname{tr}(U H_b), \qquad \|H_b - 6I\| \le C\delta, \tag{PD12.50}$$

up to an additive bath constant. At $H_b = 6I$ it has a unique minimum at I, a positive local Hessian, and a positive gap away from any sufficiently small fixed identity chart. Compactness, the implicit function theorem, and a smaller fixed δ give the same properties uniformly for all good baths, with the minimum at $x_*(\mathfrak{b})$ inside the chart. The local Hessian is bounded above and below there.

Direct integration after subtracting the minimum action bounds the conditional normalizer between $c\beta^{-d/2}$ and $C\beta^{-d/2}$, where $d = \dim G$. The contribution outside the chart is exponentially small, and the local Gaussian upper and lower integrals give the powers. Therefore, in

$$z = \sqrt{\beta}\,(x - x_*(\mathfrak{b})), \tag{PD12.51}$$

the normalized one-link density is bounded below by $c > 0$ on a fixed unit ball, uniformly in $\beta \geq \beta_0$ and all good baths.

Set $H(z) = \beta G_{H,b}(x_* + \beta^{-1/2} z; \mathfrak{b})$. The Hessian lower bound of Section 6.3 gives $D_z^2 H \geq cI$. Choose $\varphi \geq 0$, $\varphi \in C_c^\infty$ of the ball, with $\int \varphi = 1$. For every scalar m,

$$dc \leq \int (\Delta H)\varphi = \int (H - m)\Delta\varphi \leq \|H - m\|_{L^2} \|\Delta\varphi\|_{L^2}. \tag{PD12.52}$$

Combining this lower bound on the L^2 distance to every constant with the conditional density lower bound yields

$$\mathrm{Var}_{b,\beta}(G_{H,b} \mid \mathfrak{b}) \geq c_1 \beta^{-2} \quad \text{on every good bath.} \tag{PD12.53}$$

No gradient at the conditional minimum was dropped. In particular, the argument remains valid if the linear term in the rescaled H is large. This is a nonlinear finite-dimensional estimate, not a Gaussian weak-convergence assertion.

155.6.5 Completion and a fixed physical insertion

Equations PD12.44, PD12.46, PD12.47, and PD12.53 give

$$\mathrm{Var}(G_H) \geq c\beta^{-2} n^4. \tag{PD12.54}$$

Multiplying by the square of the actual normalization in PD12.39 proves

$$\mathrm{Var}(O_H(1)) \geq c g^{-4} a^{-4} \beta^{-2} n^4 = c c_\beta^{-2} a^{-4} n^4 = c' T^4 a^{-8}. \tag{PD12.55}$$

For a fixed bounded real-valued physical test f equal to one on a box Q of positive physical volume, choose all free links in an interior sub-box. Its collar contains every determining bond and every shifted pointing used by the fixed symmetrization. Then every H summand which varies with one selected free link has test value exactly one, so PD12.49 is unchanged. There are at least $c|Q|a^{-4}$ selected links for small enough a. The real values of f outside these determining neighborhoods can be arbitrary. Hence

$$\mathrm{Var}_{\mu_\beta}(O_H(f)) \geq c_f a^{-8}, \qquad \beta \geq \beta_0, \tag{PD12.56}$$

where $c_f > 0$ depends on the plateau volume and fixed local geometry. The result is not restricted to a source filling the whole torus.

For a complex test the identical-label connected coefficient is the bilinear quantity $\mathbb{E}[(O_H - \mathbb{E}O_H)^2]$, and it need not be nonnegative. The real-test hypothesis is therefore essential to PD12.56 as stated. Pairing a complex test with its conjugate gives the Hermitian variance instead; that is a different two-label pairing.

Centering $O_H(f)$ by its genuine single-source mean does not change PD12.55–56. Supplying a mixed-source subtraction of exactly that variance would change the original source prescription and must be displayed if such a renormalized coincident product is chosen.

155.7 Absolute scalar masses and the remaining upper-bound problem

The terminal mixed logarithmic scalar in PD12.38 is nonnegative and satisfies the lower bounds above. Against the raw measure its connected numerator is

$$C_{12}^{\mathrm{raw}} = N_{12} - \frac{N_1 N_2}{N_0} = Z_\beta \operatorname{Var}_{\mu_\beta}(O_H(f)), \qquad N_0 = Z_\beta > 0. \tag{PD12.57}$$

Thus its absolute mass is exactly $Z_\beta \operatorname{Var}(O_H(f))$. A genuinely carried, independent field-free factor e^s multiplies that mass by $|e^s|$. No lower bound for the raw Z_β, and no source-dependent replacement of Z_β, is being inserted. Likewise the raw mass of the contact in PD12.30 is $Z_\beta \mathbb{E}|b_{HH}|$, with the same original absolute scalar prefactors.

For the uniform H source, the identity PD12.16 reads

$$\operatorname{Var}(O_H) = \operatorname{Var}(r_H) + \mathbb{E} b_{HH}. \tag{PD12.58}$$

The contact estimate PD12.30 controls the second term in absolute value. It gives no matching upper bound for $\operatorname{Var}(r_H)$, which is a genuine ordinary connected source. r_H contains directional forces, and their covariance is not the variance of a complete total current. The exact Ward upper bound for WC11's current square therefore cannot be copied to the entire left side of PD12.58.

A fully elementary but weaker bound can be proved without a mixing hypothesis. Since

$$0 \le G_H \le C_N S_0, \tag{PD12.59}$$

and

$$\mathbb{E}_\beta e^{\beta S_0/2} = \frac{Z_{\beta/2}}{Z_\beta} \le e^{A_\beta}, \qquad A_\beta = C \# E \log(2+\beta), \tag{PD12.60}$$

Markov's inequality followed by integration of the tail gives

$$\mathbb{E} S_0^2 \le C \beta^{-2} (A_\beta + 1)^2. \tag{PD12.61}$$

For example split the integral $\int_0^\infty 2t\, \mathbb{P}(S_0 > t) dt$ at $t = 2A_\beta/\beta$, and use $\mathbb{P}(S_0 > t) \le e^{A_\beta - \beta t/2}$ thereafter. It follows that

$$\operatorname{Var}(O_H(1)) \le \mathbb{E} O_H(1)^2 \le C c_\beta^{-2} a^{-4} [\# E \log(2+\beta) + 1]^2. \tag{PD12.62}$$

With $\# E = 4(T/a)^4$, its leading size is $C T^8 a^{-12} \log^2(2+\beta)$, not the desired $C T^4 a^{-8}$. It is recorded only as a verified finite-volume upper bound. It does not close the marked induction.

For a bounded test supported on an actual completed set Y_f, let $N_f = \# Y_f$ denote its plaquette count. Then $|G_H(f)| \le C \|f\|_\infty \sum_{p \in Y_f} s_p$. Compactness and PD12.31 also give the support-count bound

$$\operatorname{Var}(O_H(f)) \le C c_\beta^{-2} \|f\|_\infty^2 a^{-4} \beta N_f^2 \log(2+\beta). \tag{PD12.63}$$

One can take the minimum of PD12.63 and the global bound PD12.62 times $\|f\|_\infty^2$. Neither is a substitute for a connected bound proportional to N_f. A uniform spectral gap, a summable covariance

estimate for this ordinary sector, or an appropriate all-history marked cluster estimate would be additional substantive inputs; none is asserted here.

155.8 Original meshes, gains, records, and conclusions

The explicit birth calculation PD12.20–30 is on one original lattice at spacing a, coupling g, and Wilson density ρ. It does not replace the original mesh of an inherited contact word by the current output mesh. If a primitive label carries a genuine field-independent normalization or saved scale factor Γ_i, multiply its whole O_i, q_i, t_i, r_i by that factor. Every bilinear coefficient and every bound then carries $\Gamma_1\Gamma_2$. A factor depending on live fields is differentiated and requires its actual jets; it cannot be treated as a saved numerical gain.

For labels born on different original meshes, the general formulas PD12.11–18 and PD12.33–37 retain their original vector fields and readouts. The common-mesh force formula PD12.22 is not applied by substituting a convenient later a. A nonlinear prepared readout must first be pulled back with its actual derivative and constraint data, and the inherited density uses its complete effective score. Those are precisely the additional estimates not supplied by the bare computation.

The selector alternatives, genuine native denominators, reference coordinates, component multiplicities, and all old marks remain in the map $\mathcal{T}$ of PD12.36. Kernel response cancellation is used only after the complete selector sum and complete native integral for which $\mathsf{K1} = 1$. It is not applied to an individual rare component or before the old mark's prescribed integration. A boundary current is kept as part of the full weak derivative; no product of two unrelated boundary distributions is postulated.

The result of this calculation can be summarized precisely.

Obligation	Result obtained here	Remaining issue
Identify the original mixed-source action	Bare $A_{12} = 0$; actual output is PD12.37 from the joint physical product	Quantitative control on every selected mixed output
Keep mixed current–ordinary compensation	PD12.7, PD12.10, PD12.13, and the explicit H formula PD12.22	Identification with a specified all-history projection and its effective-score jets
Retain scalar companions and kernel responses	PD12.8, PD12.10, PD12.29, and complete weak rules PD12.33–35	Boundary and internal-derivative bounds on the actual selected support
Bound an actual physical contact at birth	One-reserve bound PD12.27 and absolute mass PD12.30	Complex common-domain and inherited transition bounds
Test complete scalar cancellation in the physical prescription	Actual H variance lower bound PD12.40 and PD12.56; total scalar cancellation is impossible for this raw coincident source	Correct quantitative lower/contact filing and any explicitly prescribed renormalization
Obtain matching full scalar upper bound	Elementary bounds PD12.62–63 only; contact part has the sharper PD12.30	Summable ordinary covariance / all-history marked estimate

The nonzero H variance is not a contradiction to MT36, whose state explicitly retains lower and contact sectors. It proves that those sectors cannot be omitted or declared canceled merely because a current has zero mean or an auxiliary reference kernel has unit mass. The simultaneous current,

ordinary, contact, and scalar filing must reproduce PD12.38 and PD12.57 with their original absolute masses.

The new current–ordinary formula closes one concrete bare bookkeeping and real-energy estimate. The complete same-measure all-history MT36 estimate remains unproved, because the required inherited ordinary covariance, fine internal derivatives, boundary responses, and actual mixed-domain admission have not been bounded here. No assertion in this note closes the Yang–Mills existence proof.

156.1 Two physical marks and the complete compensated coefficient

Fix a finite positive Wilson measure with its absolute mass retained,

$$d\mu = \rho\, dU, \qquad \mu(1) = Z. \tag{DT12.1}$$

On the compact fine configuration space let V_i be smooth vector fields, including their actual source weights and original mesh factors. The density adjoint is $D_V = -\operatorname{div}(V\,\cdot)$. Suppose

$$t_i = \rho^{-1} D_{V_i} \rho, \qquad O_i = t_i + r_i, \qquad i = 1, 2. \tag{DT12.2}$$

This is a decomposition of the physical mark, not a prescription for independently sourcing its two terms. For two distinct squarefree labels, keep the actual source action

$$A = \eta_1 O_1 + \eta_2 O_2 + \eta_1 \eta_2 A_{12}. \tag{DT12.3}$$

At the bare XP3 source, A_{12}=0. At an inherited step it is the coefficient actually generated by the preceding joint integrations.

Set $\mathcal{V} = \eta_1 V_1 + \eta_2 V_2$. Expanding the exact WC11 change of density, with the order of the vector fields retained, gives

$$\mathcal{B}_{\mathcal{V}}(A) = \eta_1 r_1 + \eta_2 r_2 + \eta_1\eta_2 b_{12}, \tag{DT12.4}$$

where

$$\boxed{b_{12} = A_{12} + \tfrac{1}{2}(V_1 t_2 + V_2 t_1) + V_1 r_2 + V_2 r_1.} \tag{DT12.5}$$

Indeed $e^{\mathcal{V}}A$ contributes $A_{12} + V_1(t_2 + r_2) + V_2(t_1 + r_1)$, whereas $\int_0^1 e^{s\mathcal{V}} t\, ds$ contributes $\frac{1}{2}(V_1 t_2 + V_2 t_1)$ at this degree. Every derivative of a field-dependent coefficient is included.

The defect $\mathcal{R}_{12}$ of WC11.46 is the first two terms of DT12.5. It is the complete degree-two compensation only when both ordinary residuals r_i vanish. For the physical H source they do not vanish. Its mixed current–ordinary terms are mandatory.

156.2 The joint-kernel formula, before division

Let $K(U, d\xi)$ be the literal source-free auxiliary kernel. Its density or distribution includes every constraint, selector, coordinate Jacobian, live reference, effective action and genuine denominator. Let $\Psi(U, \xi)$ be the actual carrying test. The degree-two coefficient of its original marked integral is exactly

$$\begin{aligned}[\eta_1\eta_2] \int d\mu\, e^{A} K\Psi = \int d\mu\, \{&(b_{12} + r_1 r_2) K\Psi \\ &+ r_1 V_2(K\Psi) + r_2 V_1(K\Psi) \\ &+ \tfrac{1}{2}(V_1 V_2 + V_2 V_1)(K\Psi)\}.\end{aligned} \tag{DT12.6}$$

This follows by expanding $\int d\mu\, e^{\mathcal{B}_{\mathcal{V}}(A)} e^{\mathcal{V}}(K\Psi)$. The expressions $V_i(K\Psi)$ and $V_i V_j(K\Psi)$ are derivatives of the whole tested kernel. They are not products of separately defined delta-boundary distributions.

If the actual extraction also singles out a degree-two current operator L_{12}, with $t_{12}\mu = L_{12}^{*}\mu$, its contribution is retained by replacing b_{12} with $b_{12} - t_{12}$ and adding $L_{12}(K\Psi)$ in DT12.6. This remains true for an ordered higher differential word: at source degree two its coefficient $\eta_1\eta_2$ has no product with another nonempty source coefficient. No derivation property of L_{12} is asserted. For its interpretation as a current require $L_{12}1 = 0$; otherwise its zeroth-order scalar is retained separately. Its original word, mesh, normalization and component multiplicity remain attached.

For a genuinely normalized complete kernel, K1=1 as a function of every old variable that the directions vary. Summing all selector alternatives and integrating the entire new fiber then gives

$$(V_i K)1 = V_i(K1) = 0, \qquad [(V_i V_j + V_j V_i)K]1 = 0. \tag{DT12.7}$$

These equalities concern the full kernel. A single selected branch K_ω generally satisfies neither of them. If the carrying function is not an old function annihilated by the relevant directions, its response terms in DT12.6 also remain.

At the terminal old test 1, compact integration by parts and DT12.6 give the prescription-preserving scalar identity

$$\boxed{\mathbb{E}A_{12} + \mathrm{Cov}(O_1, O_2) = \mathbb{E}b_{12} + \mathrm{Cov}(r_1, r_2).} \tag{DT12.8}$$

The expectations here divide once by the original Z. In raw masses the equality is multiplied by Z; an independent old scalar multiplies both sides by its own absolute factor.

DT12.8 makes two distinct cancellations visible. The auxiliary normalization cancels its own complete kernel responses. The original physical source still has the scalar on the right. Setting it to zero requires an additional identity or estimate for those actual terms.

156.3 The degree-two defect after a smooth conditional integration

The following statement applies on an actual smooth conditional chart, with differentiable conditional means and all displayed products integrable. Those are checked hypotheses of this statement, not assumptions asserted for every mixed history.

Let q be the actual observation and $\nu = q_*\mu$. Put

$$v_i = Dq\, V_i, \quad \bar{v}_i = \mathbb{E}(v_i \mid q), \quad \bar{t}_i = \mathbb{E}(t_i \mid q), \quad \bar{r}_i = \mathbb{E}(r_i \mid q). \tag{DT12.9}$$

Here the full current identity DT12.2 holds on the realization being conditioned. In particular, if an auxiliary kernel has already been attached, its effective score must be included before this hypothesis is used. The pushed current identity is

$$\bar{t}_i\nu = D(\bar{v}_i\nu), \qquad D = -\operatorname{div}. \tag{DT12.10}$$

No coordinate of a live reference is excluded from q when V_i varies it.

Exact finite-source conditioning gives

$$a_i = \bar{t}_i + \bar{r}_i, \qquad a_{12} = \mathbb{E}(A_{12} \mid q) + \operatorname{Cov}(O_1, O_2 \mid q). \tag{DT12.11}$$

Thus the coarse compensated coefficient is

$$\begin{aligned} b_{12}^c = a_{12} &+ \tfrac{1}{2}(\bar{v}_1 \cdot \nabla\bar{t}_2 + \bar{v}_2 \cdot \nabla\bar{t}_1) \\ &+ \bar{v}_1 \cdot \nabla\bar{r}_2 + \bar{v}_2 \cdot \nabla\bar{r}_1. \end{aligned} \tag{DT12.12}$$

It obeys

$$\boxed{b_{12}^c\nu = \mathbb{E}(b_{12} \mid q)\nu + \operatorname{Cov}(r_1, r_2 \mid q)\nu + D(J_{12}\nu),} \tag{DT12.13}$$

where

$$\begin{aligned} J_{12} = \tfrac{1}{2}\{&\operatorname{Cov}(t_2, v_1 \mid q) + \operatorname{Cov}(t_1, v_2 \mid q)\} \\ &+ \operatorname{Cov}(r_2, v_1 \mid q) + \operatorname{Cov}(r_1, v_2 \mid q). \end{aligned} \tag{DT12.14}$$

Proof. For any smooth fine scalar f, the product rule in the complete fine identity and the pushed identity DT12.10 gives

$$\operatorname{Cov}(f, t_i \mid q)\nu = D\big(\operatorname{Cov}(f, v_i \mid q)\nu\big) + \{\mathbb{E}(V_i f \mid q) - \bar{v}_i \cdot \nabla\mathbb{E}(f \mid q)\}\nu. \tag{DT12.15}$$

Apply this once to each mixed pair $f = r_j$. Apply it to $f = t_j$ in both orders and average. Expand $\operatorname{Cov}(t_1 + r_1, t_2 + r_2 \mid q)$ in DT12.11. The retained derivatives cancel those in DT12.12, leaving exactly DT12.13. This proof uses neither Gaussianity nor commuting V_i.

For a purely current pair r_1=r_2=0, DT12.13 becomes

$$\boxed{\mathcal{R}_{12}^c\nu = \mathbb{E}(\mathcal{R}_{12} \mid q)\nu + \tfrac{1}{2}D([\operatorname{Cov}(t_2, v_1 \mid q) + \operatorname{Cov}(t_1, v_2 \mid q)]\nu).} \tag{DT12.16}$$

Even a fine defect equal to zero can therefore acquire a nonzero coarse current companion. Discarding that companion would break the next transport identity. If each v_i is literally a function of q, the covariance companion vanishes; projectability must be proved on the actual realization.

156.5 Quantitative meaning and the remaining model-specific step

On the admitted smooth chart, the explicitly represented form DT12.13 has the bound

$$\mathfrak{M}(b_{12}^{c}\nu) \le \int |b_{12}|\, d\mu + \int |\operatorname{Cov}(r_1, r_2 \mid q)|\, d\nu + \int |J_{12}|\, d\nu. \tag{DT12.21}$$

All allowed target-coordinate components are counted in the last norm. Source weights and canonical word weights belong in the coefficient measures before this bound is used. It does not insert a factor of the current mesh into an old source word. Here $\mathfrak{M}$ is the represented-distribution norm of CT9 and SC11, with the indicated actual target words. It is not ordinary total variation of the scalar measure on the left: the variation of $D(J_{12}\nu)$ is not bounded merely by $\int |J_{12}|\, d\nu$.

Equation DT12.21 is not claimed to be the optimal bound: SC11 can further cancel the complete conditional-covariance packet using internal derivatives. The new information here is which original degree-two coefficient and which current–ordinary responses that cancellation must actually transport. NB12 gives quantitative internal and sharp-boundary estimates for admitted native reference primitives. Applying them to every inherited physical current still requires the actual mixed chart, the correct lift of its directions, the bounds for its inherited density multiplier, and the original saved grade.

For the H insertion, the absolute estimate for its explicit compensated birth in PD12 does not estimate the remaining $\operatorname{Cov}(r_1, r_2 \mid q)$ by itself. Nor does the latter admit removal by the closed normalization DT12.7. This is the specific ordinary connected row still left in that decomposition.

157.1 The native coordinates, mass and analytic input

Fix the complete old datum b, including all still-live reference variables. Use one original allowed relative forest and retain its actual q free group coordinates. If m is the group dimension, then

$$n = mq, \qquad q = |E| - |V| + |P| \tag{NB12.1}$$

when the native record has the rooted forest form of RV11.13. No comparison with a different record is made in NB12.1. If q=0 there is no internal derivative or cutoff boundary.

The genuine reference denominator has the form

$$D(b) = P(b)\widehat{D}(b), \qquad \widehat{D}(b) = \int_{\Omega_R} f(B;b)\,dB, \qquad R = M_0 p_0(g), \tag{NB12.2}$$

where

$$\Omega_R = \{B : |B_e|_* < R \text{ for every actual free coordinate } e\}. \tag{NB12.3}$$

The norm $|\cdot|_*$ is the norm in the original logarithmic selector. It is not replaced by a new cutoff. Its fixed-dimensional unit ball has a Lipschitz boundary, and fixed norm-equivalence constants are retained. Corners of the product have zero surface measure. The prefactor P contains the original classical action, the actual $n \log g$, and the original Haar normalization.

On the admitted real primitive, write the exact native density as

$$f(B) = \exp\{-\tfrac{1}{2}BQB + \ell \cdot B + W_{\mathrm{nl}}(B)\}\, J(gB). \tag{NB12.4}$$

Here Q is the native axial quadratic form; ℓ is the actual weighted linear remainder; W_{nl} is the evaluated nonlinear action remainder, including its factor g^{-2}; and J is the actual product Haar Jacobian divided by its value at zero. Changing a sign convention for the remainder only changes the sign used in its definition in NB12.4.

The primary inputs used here are precise. R-II, printed pp.356–358, (1.2)–(1.9), supplies the positive axial quadratic form, the bounded linear term by stationarity and separation, and the small third-order nonlinear term. Its component is contained in a current cube of side 100M. The target response is the analytic minimum of the 1985 variational paper, Proposition 9, (172)–(190); Proposition 4, (97)–(98), supplies its analytic nonlinear action derivative. See the R-II reference expansion and the inspected 1985 variational paper.

For sufficiently small g with fixed M, the original target range $|gB|_* \le gR = M_0\varepsilon_k$ lies inside that admitted reference chart. Its fixed finite enlargement $|gB|_* \le 3gR$ also lies inside the same regular

response theorem, after decreasing the small-coupling ceiling. This is a statement about the reference primitive with a uniformly regular real background. It is not a claim about simultaneous preparation walls of an arbitrary old mixed coefficient. The integration domain remains exactly NB12.3.

The native cubic estimate can be used on this enlarged complex target range by rerunning its norm bound with $3M_0$ in place of M_0. This does not follow from analyticity alone. The action remainder being estimated is the one displayed in the 1987 small-field paper, (2.6)–(2.8), printed p.266: it includes the nonlinear-minimum and constraint corrections. Its Section 3, printed pp.269–270, explicitly treats the primitive as analytic in the fluctuation variable with a small quadratic majorant. The 1985 Proposition 4 bounds the action derivative on the complex regularity ball, and Proposition 9 supplies the complex response in that ball. Thus the absolute norm estimates, including the summed native $|\Lambda|$ bound of R-II (1.2), hold after this fixed finite target enlargement with changed constants. See the 1987 expansion and analytic primitive.

In the weaker form already sufficient here, for any one fixed $0 < \delta < 1$,

$$\sup_{\{|B_e|_* \le 3R\}} |W_{\mathrm{nl}}(B)| \le \epsilon_{\mathrm{nl}}(g), \qquad \epsilon_{\mathrm{nl}}(g) = C_{M,\delta}|\Lambda|_{\mathrm{nat}} g^{\,1-\delta}. \tag{NB12.5}$$

The fixed finite number $|\Lambda|_{\mathrm{nat}}$ is the actual current-scale component size of (1.2), bounded by CM^4. It is not the number of microscopic sites in Z. NB12.5 uses the summed native action estimate of (1.2), not an inference from the number n of coarse coordinates. Analyticity alone would not bound that summed action.

The same native expansion gives

$$\lambda_M I \le Q \le \Lambda_M I, \quad |\ell| \le C_M, \quad e^{-C_M} e^{-\frac{1}{2}BQB} \le f(B) \le e^{C_M} e^{-\frac{1}{2}BQB} \quad (B \in \overline{\Omega_R}). \tag{NB12.6}$$

These real bounds include the actual linear remainder, the nonlinear term and J. They are the RV11.4 bounds on this admitted sector.

157.2 Cauchy estimates in the actual internal directions

For each fixed multi-index α, Cauchy's estimate on the enlarged target range gives

$$|\partial_B^\alpha W_{\mathrm{nl}}(B)| \le C_{M,\alpha}\epsilon_{\mathrm{nl}}(g) R^{-|\alpha|} \quad (B \in \overline{\Omega_R}). \tag{NB12.7}$$

A product polydisc of radius $c_m R$ around each real B in the original cutoff lies in the complex norm range used above. The norm-equivalence constants and the actual n-coordinate multi-index are included in $C_{M,\alpha}$. No source direction has been changed into an internal B derivative in obtaining NB12.7.

The Haar Jacobian is analytic and nonzero in a fixed group logarithm chart containing $3gR$ for small g. It has no linear logarithmic term at zero. Thus

$$|\partial_B \log J(gB)| \le C_M g^2 |B|, \qquad |\partial_B^\alpha \log J(gB)| \le C_{M,\alpha} g^{|\alpha|} \quad (|\alpha| \ge 2). \tag{NB12.8}$$

All actual free-coordinate factors of J are retained. Since R tends to infinity and ϵ_{nl} tends to zero, NB12.4, NB12.7 and NB12.8 imply, at every fixed order r,

$$|\partial_B^\alpha f(B)| \le C_{M,r}(1 + |B|)^r f(B), \qquad |\alpha| \le r. \tag{NB12.9}$$

For the first logarithmic derivative the only unbounded term is $-QB$, which is linear. Higher logarithmic derivatives are bounded at each fixed order. The ordinary derivatives of f are then the finite Bell-polynomial expressions in these logarithmic derivatives, proving NB12.9.

For a coordinate vector field, the complete internal score of the actual density is

$$t_y = -\partial_{B_y} \log f = (QB)_y - \ell_y - \partial_{B_y} W_{\mathrm{nl}} - \partial_{B_y} \log J. \tag{NB12.10}$$

The nonlinear score response has the quantitative bound

$$|t_y - t_y^0| \le C_M \epsilon_{\mathrm{nl}}(g)/R, \qquad t_y^0 = (QB)_y - \ell_y - \partial_{B_y} \log J. \tag{NB12.11}$$

Thus using the native Gaussian-linear-Haar score in place of the actual score requires this explicit nonlinear response. It is not an exact equality. For a vector field with polynomially bounded coefficient jets, the analogous difference is its literal contraction with $-\nabla_B W_{\mathrm{nl}}$.

157.3 The actual selector boundary has a small weighted trace

Let $d\sigma$ denote Euclidean surface measure on the original product-norm boundary. Let $c_* > 0$ be such that $|x|_* = R$ implies $|x|_2 \ge c_* R$. For every fixed p and every sufficiently small fixed $\theta \ge 0$,

$$\boxed{\frac{1}{\widehat{D}} \int_{\partial\Omega_R} (1 + |B|)^p e^{\theta |B|^2} f(B)\, d\sigma(B) \le C_{M,p}\, q\, R^{m-1} (1+R)^p e^{-c_M R^2}.} \tag{NB12.12}$$

For example one may take $\theta \le \lambda_M/8$, reducing c_M by the fixed norm-equivalence constants.

Proof. The original cutoff contains a fixed Euclidean ball for small g. Hence

$$\widehat{D} \ge z_{M,n} > 0 \tag{NB12.13}$$

by the real lower bound NB12.6, exactly as in RV11.6. Decompose the boundary, up to a null set, into the q faces on which one group coordinate lies on $R\partial\mathcal{B}_*$ and all others lie in $R\mathcal{B}_*$. Its surface measure in that coordinate is R^{m-1} times the fixed unit surface measure. On the face the Euclidean norm of that coordinate is at least $c_* R$.

Use NB12.6 to split the Gaussian exponent into a factor $e^{-c_M R^2}$, a factor absorbing the polynomial weight, and an integrable Gaussian in all the other original coordinates. The latter integral is bounded by its whole-space Gaussian integral; the density itself has not been extended beyond its original chart. Sum the q faces and divide by NB12.13. This proves NB12.12 with n retained in the constants.

The estimate is for a boundary trace, not an inference from a bulk probability of a boundary set. That set has bulk measure zero. Continuity and the pointwise density envelope on the original wall are the additional facts used here.

157.4 All fixed-order cutoff commutators without products of deltas

Write $\chi = 1_{\Omega_R}$, and first use the literal coordinate density derivatives $D_i = -\partial_{B_i}$. For the ordered word $D_w = D_{i_r} \cdots D_{i_1}$, the exact operator identity is

$$\boxed{[D_w, \chi] f = \sum_{j=1}^{r} D_{i_r} \cdots D_{i_{j+1}} \left[(D_{i_j} \chi) D_{i_{j-1}} \cdots D_{i_1} f \right].} \tag{NB12.14}$$

An empty ordered product is the identity. The first derivative of the original indicator is the finite surface measure

$$D_i \chi = n_i\, d\sigma_{\partial\Omega_R}, \tag{NB12.15}$$

where n is the outward Euclidean unit normal. Formula NB12.14 follows by the telescoping commutator identity $[AB,\chi]=A[B,\chi]+[A,\chi]B$. It is valid for a Lipschitz domain in the distributional sense.

Each term in NB12.14 is an allowed ordered derivative of one finite boundary measure. Normals or surface measures are not multiplied by other delta distributions. Higher contact order is represented by the outer word, with its original order unchanged.

Combining NB12.9 and NB12.12 gives the quantitative represented boundary bound

$$\boxed{\mathfrak{M}_{\le r-1}\left(\widehat{D}^{-1}[D_w,\chi]f\right)\le C_{M,r}\,q\,R^{m-1}(1+R)^{r-1}e^{-c_M R^2}.} \tag{NB12.16}$$

This is a represented-distribution norm. An order-two commutator can contain a derivative of a surface measure; it is not being called an ordinary signed measure of this total variation.

For an actual smooth vector field V in these coordinates,

$$[D_V,\chi]f=(V\cdot n)f\,d\sigma_{\partial\Omega_R}. \tag{NB12.17}$$

The same telescoping formula applies to an ordered word of such density adjoints. A fixed polynomial envelope for the coefficient jets of those V's changes only the displayed polynomial power of R and its literal coefficient factor. An old physical current can use this statement only after its actual vector fields and all their kernel/reference responses have been identified in this chart.

157.6 Absolute masses, original meshes and limits of the result

All normalized trace estimates above are multiplied by the actual $\widehat{D}$ when used for the extracted unnormalized density, and by $P\widehat{D} = D$ for the raw reference mass. An independent old scalar e^s contributes its own $|e^s|$. Neither a small D nor its coordinate factor g^n has been replaced by a lower bound for D.

If a contact word was born with a coefficient $\Gamma_j(a_j, g_j)$, that same coefficient multiplies NB12.16 and NB12.21. The current coupling enters the native tail through

$$e^{-c_M R^2} = e^{-c_M M_0^2 p_0(g)^2}. \tag{NB12.24}$$

It dominates any fixed polynomial in the current inverse coupling. It does not dominate an arbitrary power of a different, older microscopic mesh at a fixed later coupling. No such birth factor has been inserted here.

The following are still missing for all-history MT36: the identification of the physical current directions with their full native kernel lifts; their coefficient-jet and coordinate-incidence bounds; the internal derivative estimates for every inherited effective multiplier, rather than the native action factor and the changed-weight case; and the simultaneous complex chart on the actual mixed selector support. NB12 proves the reference derivative and boundary estimates once that reference is actually the one being used. It does not supply those separate identifications.

158.2 The actual reference variable and the unequal records

Fix the complete old datum b, the original free-link graph, its pins, and its native relative forest. Let q be its actual number of free group coordinates, m the Lie algebra dimension, and $n = mq$. In the original logarithmic coordinates write

$$C(B) = \exp(igB), \qquad W_B = U_{k,Z}(C(B)V_\Lambda; b), \qquad W_0 = U_0. \tag{UW12.1}$$

The equality at zero is the original fixed-target identification: the targets at $C = 1$ are those of the relaxed unweighted minimizer. It is used only on a native primitive where this identification and the original cutoff admission hold. The background U_0 minimizes the unweighted variational problem. Changing the spatial weight below does not redefine either minimizer.

For two original spatial weights ζ_0, ζ_1, put

$$\begin{aligned} D_i &= \int dC\, \delta_G(C)\chi_G(C) \exp\{-g^{-2}A(\zeta_i, W_C)\}, \\ P_i &= \exp\{-g^{-2}A(\zeta_i, U_0) + c^{\mathrm{coord}}\}, \qquad \widehat{D}_i = P_i^{-1} D_i, \qquad I_i = \log \widehat{D}_i. \end{aligned} \tag{UW12.2}$$

The genuine coordinate scalar c^{coord} includes $n \log g$, the original Haar constants, and every other original extraction constant. It is equal here because the graph, pins, and coordinates are equal; it has not been suppressed. The original constraints, selectors, and Haar density are the same functions in both integrals. No integration by parts, selector deformation, or new gauge-delta removal occurs.

Define the actual centered action difference

$$w(B) = g^{-2}\{A(\Delta\zeta, W_B) - A(\Delta\zeta, U_0)\}, \qquad \Delta\zeta = \zeta_1 - \zeta_0. \tag{UW12.3}$$

Then the exact identities, with the native base probability K_0, are

$$\begin{aligned} I_1 - I_0 &= \log K_0(e^{-w}), \\ \log D_1 - \log D_0 &= -g^{-2}A(\Delta\zeta, U_0) + I_1 - I_0, \\ \frac{dK_1}{dK_0} &= \frac{e^{-w}}{K_0(e^{-w})}. \end{aligned} \tag{UW12.4}$$

UW12.4 alone is the previously available centered-ratio algebra. The new issue is proving a suitable bound for its actual w, without replacing it by an unspecified sensitivity input.

158.4 Integration of the score: a moment bound instead of a cutoff supremum

Use the admitted native extraction of RV11.4:

$$f_0(B) = e^{-\frac{1}{2}\langle B, Q_0 B\rangle} u_0(B), \quad \lambda_M I \le Q_0 \le \Lambda_M I, \quad e^{-C_M} \le u_0 \le e^{C_M}. \tag{UW12.12}$$

The original cutoff contains a fixed unit ball for sufficiently small running coupling, as follows from its verified native radius $R_\chi = M_0 p_0(g)$. Let v_n be the volume of that unit ball. The following explicit moment majorant keeps the coordinate dimension:

$$\begin{aligned} \widehat{D}_0 &\ge v_n e^{-C_M - \Lambda_M/2}, \\ K_0[(1+|B|)^p e^{t|B|^2}] &\le \mathfrak{M}_{p,n}(t) \\ &:= \frac{e^{2C_M + \Lambda_M/2}}{v_n} \int_{\mathbb{R}^n} (1+|B|)^p e^{-(\lambda_M/2 - t)|B|^2}\, dB, \qquad t < \lambda_M/2. \end{aligned} \tag{UW12.13}$$

The zero-coordinate case is interpreted directly: the fluctuation integral is a single value and $w = 0$. For $n > 0$, the first line is the original extracted denominator lower bound on its unit ball. Thus the second line has no inverse raw classical mass.

158.4.1 Proposition UW12.14: unequal-weight primitive comparison

For the fixed native component there are constants $c_M, C'_M > 0$ such that, if $\delta = a + b \le c_M$, then

$$\begin{aligned} |I_1 - I_0| &\le C'_M \delta, \\ \|K_1 - K_0\|_{\mathrm{TV}} &\le C'_M \delta, \\ \left| \log \frac{D_1}{D_0} + g^{-2} A(\Delta\zeta, U_0) \right| &\le C'_M \delta. \end{aligned} \tag{UW12.14}$$

The constants can be read from the moments in UW12.13 and therefore retain their dependence on the actual coordinate dimension. They may be bounded uniformly over the original bounded native reference components, $n \le n_M$. This last simplification does not bound an arbitrary union of components by the constant for one component.

Proof. Choose fixed $\sigma > 0$ with $2\sigma < \lambda_M/4$. For $b \le \sigma$, Young's inequality and UW12.10 give

$$|w| \le a|B| + b|B|^2 \le a^2/(4\sigma) + 2\sigma|B|^2. \tag{UW12.15}$$

The elementary bound $|e^{-w} - 1| \le |w| e^{|w|}$ and UW12.13 therefore imply

$$K_0 |e^{-w} - 1| \le e^{a^2/(4\sigma)} \{ a\mathfrak{M}_{1,n}(2\sigma) + b\mathfrak{M}_{2,n}(2\sigma) \} \le C'_M \delta. \tag{UW12.16}$$

Choose c_M so the last quantity is at most $1/2$. Then $|\log K_0(e^{-w})| \le 2C'_M\delta$. Subtracting the denominator from the numerator in the last line of UW12.4 gives the total variation estimate, and the raw-log estimate follows from the exact middle line of UW12.4. This proves the proposition. No estimate of $|B|$ by $\sqrt{n}\, R_\chi$ was made. **QED.**

For bounded native dimension the unweighted first and second moments can alternatively be displayed as constants times $\sqrt{n}$ and n. In that notation the distance contribution has the useful form

$$C_M \omega n \{ (\varepsilon_{\mathrm{bg}}/g) e^{-cD} + e^{-2cD} \}. \tag{UW12.17}$$

The exact, dimension-explicit expression is UW12.16 with UW12.11. The factor $\varepsilon_{\mathrm{bg}}/g$ is real background strength in the original coordinates. It must not be replaced by one unless the actual background bound justifies that replacement. When the native background is bounded by a fixed power of $p_0(g)$ times g, only that background power remains. A second factor $p_0(g)$ from a fluctuation cutoff supremum has been removed.

UW12.17 uses the verified top-scale separation hypothesis of UW12.11. The general mixed-shell formula remains UW12.9 and UW12.16, with its literal v_Q, σ_Q, and cell sums.

158.4.2 The absolute scalar excess and its sign are retained

The actual interpolated weights give

$$I(t) - I(0) = \log K_0(e^{-tw}), \quad I'(t) = -K_t w, \quad I''(t) = \mathrm{Var}_{K_t}(w) \ge 0. \tag{UW12.18}$$

These identities follow by differentiating the finite original integral. The first-order scalar term is therefore

$$-g^{-2} \sum_p \Delta\zeta(p) \{ K_0[s(P_p(W_B))] - s(P_p(U_0)) \}. \tag{UW12.19}$$

The cumulant remainder is $O_M(\delta^2)$ by the same moment proof with $|w|^2e^{|w|}$. It is an actual excess-energy expectation, not zero by normalization. A vanishing centered kernel on constants does not set UW12.19 to zero.

There is also a controlled complex interpolation in this weight parameter alone. Replace w by zw in UW12.16. For $|z|\delta < c_M$, the denominator stays nonzero and the original logarithm is analytic. Cauchy's estimate gives

$$|I^{(r)}(0)| \le r!C_M(C_M\delta)^r. \tag{UW12.20}$$

These are derivatives with respect to the spatial-weight interpolation parameter. They are not source/contact derivatives, and they confer no new source grade or mesh factor. A complex outer field comparison still needs the actual common outer chart and its separate curvature control. RV11's real-to-complex propagation can be used on such a verified chart; UW12.20 does not create that chart.

158.4.3 Actual internal derivatives of this nonconstant multiplier

There is a useful additional consequence for the internal-current problem. UW12.8 comes from the uniform first derivative itself:

$$\|\partial_{B_y}F_B\|_{2,Q} \le gK(Q,y). \tag{UW12.20a}$$

Differentiating the exact real Wilson energy, and then using $\|F_B\|_{2,Q} \le \varepsilon_Q + g\sum_z K(Q,z)|B_z|$, gives

$$|\partial_{B_y}w(B)| \le \alpha_y + \sum_z H_{yz}|B_z|, \qquad \|\nabla_B w(B)\|_2 \le a + 2b|B|. \tag{UW12.20b}$$

Gauge-frame derivatives which merely conjugate a plaquette have zero pairing with the derivative of its gauge-invariant Wilson energy. The physical target derivative and all its non-gauge terms remain in UW12.20a. Thus no omitted gauge-frame current is used to obtain UW12.20b.

Consequently the actual multiplier $h(B) = e^{-w(B)}$ satisfies

$$\|\nabla_B h(B)\|_2 \le |h(B)|(a + 2b|B|). \tag{UW12.20c}$$

Every fixed native weighted moment of this derivative is $O_M(a+b)$ by UW12.13–16, with the original carrier degree and reserve retained. This is a proved internal derivative for a nonconstant Wilson multiplier; it is stronger than assuming an arbitrary inherited multiplier has a small internal derivative. It is a derivative in the actual auxiliary variable B. It does not identify an old physical fine-link current with a vertical derivative in B.

Higher internal derivatives require the original analytic margin, or a separately proved differentiated variational equation. On a general mixed domain, if the actual simultaneous complex B domain contains the Euclidean ball of radius r_B about the point, and the same complex curvature/target-response estimates hold there, Cauchy's formula for the first derivative gives

$$\|\nabla_B^2 w(B)\|_{2\to 2} \le C_M\{a/r_B + b(1 + |B|/r_B)\}. \tag{UW12.20d}$$

To see this without a hidden coordinate count, apply one-variable Cauchy to $D_v w(B + zu)$ for unit vectors u, v; the gradient bound holds on the actual contained ball. Its radius is a distance in the

native B coordinates, not an asserted physical-source radius. If r_B is bounded below, UW12.20d supplies the corresponding graded second derivative of h, using $D^2h = h\big[(Dw)^{\otimes 2} - D^2w\big]$. This radius hypothesis is not proved for an arbitrary mixed history by real smoothness or by UW12.20b.

For the verified top-scale native branch there is a stronger, model-specific conclusion. The actual reference target enlargement proved in NB12.5 keeps the same fixed-target map on $|B_e|_* \le 3R$, $R = R_\chi$, after reducing the small-coupling ceiling. It follows from the native target range and the variational paper's (172)–(190), Proposition 9; no old mixed wall is enlarged. On this complex reference branch use the holomorphic Wilson action

$$s_{\mathbb{C}}(P) = N - \tfrac{1}{2}\operatorname{tr}(P + P^{-1}), \qquad |ds_{\mathbb{C}}(P)[\dot P]| \le C\|P - I\|\|\dot P\|. \tag{UW12.20g}$$

The last estimate holds on the actual small plaquette chart. The complex localized response from Proposition 9 and the retained top-scale curvature bounds therefore give $\|D_B w(z)\| \le C_M\big[a + b|z|\big]$ on that same enlarged range. A real point of the original cutoff has a contained Euclidean complex B ball of radius $c_m R$; the fixed norm-equivalence constants are retained in c_m. Applying Cauchy to the first derivative on that ball proves, for each fixed $r \ge 2$,

$$\|D_B^r w(B)\| \le C_{M,n,r}\left\{\frac{a}{R^{r-1}} + bR^{2-r}\left(1 + \frac{|B|}{R}\right)\right\}. \tag{UW12.20h}$$

This is an actual internal margin in the native reference coordinates. The complex response estimate is used in addition to the analytic action-remainder estimate; an absolute bound on the remainder alone would not prove its localized small factor.

Combining UW12.20b,h with the finite derivative formulas for $h = e^{-w}$, and using $R \ge 1$, gives the quantitative native jet moments

$$\sum_{1 \le |\alpha| \le r} K_0\big[(1 + |B|)^p e^{\theta|B|^2}|\partial_B^\alpha h|\big] \le C_{M,n,r,p,\theta}(a + b). \tag{UW12.20i}$$

Here $a + b$ is sufficiently small, r, p are fixed, and θ leaves the same Gaussian margin used in UW12.15. The multi-index sum retains the actual n and its finite incidence factors. Each term in a derivative of h contains at least one derivative of w; products of two or more such derivatives remain bounded by a constant times $a + b$ because $a + b \le 1$. The remaining polynomial factors are integrated with UW12.13. Thus this actual changed-weight multiplier has the internal second derivatives needed by the degree-two conditional-covariance algebra on this native sector. Identifying an old physical current's complete lift and its observable/kernel responses is still a separate obligation.

There is no need to discard a cutoff current to use the first derivative. For the original real cutoff domain Ω, density f_0, and an actual auxiliary vector field v, the complete current distribution is

$$-\operatorname{div}(v f_0 1_\Omega) = [-\operatorname{div}(v f_0)]1_\Omega + (v \cdot n) f_0 \, d\sigma_{\partial\Omega}. \tag{UW12.20e}$$

Its pairing with h is exactly $\int_\Omega (v \cdot \nabla h) f_0$; its pairing with one is zero. In a closed reference fiber this supplies the quantitative complete current cancellation through UW12.20c. The original cutoff-boundary partner stays in UW12.20e. For polynomially bounded v, the native product cutoff has q free group-coordinate boundary faces. The fixed Lie algebra norm ball has surface measure at most $C_m R^{m-1}$; nonsmooth intersections are interpreted by its finite-perimeter boundary measure. UW12.12–13 then give

$$\frac{1}{\widehat{D}_0}\int_{\partial\Omega_R}(1+|B|)^p e^{\theta|B|^2} f_0\, d\sigma \le C_{M,p,\theta}\, q(1+R)^{p+m-1}e^{-c_M R^2}, \qquad \theta < \lambda_M/4. \tag{UW12.20f}$$

Indeed one block lies on its radius-R boundary, the remaining blocks are integrated with the original Gaussian majorant, and the original extracted denominator is bounded below as in UW12.13. Fixed norm equivalences affect c_M. The actual face count q has not been replaced by one. This bounds the partner in UW12.20e and its centered $h - K_0 h$ version by a polynomially weighted $e^{-c_M R_\chi^2}$ tail; the centered version retains the factor $a + b$, with two extra harmless polynomial powers. It uses $|h-1| \le |w|e^{|w|}$ before the boundary estimate. An output depending on B still requires the full conditional observable/kernel responses and the boundary traces of those responses; they have not been inferred from a closed-fiber identity.

158.5 Complete carriers and the one available reserve

Let F_i be the complete original carrying functions on this fiber. They may contain old integrations, current partners, retained readouts, and independent scalar factors. Suppose a literal conditional envelope is available at the fixed full old state:

$$\|F_0(B) - A_0\| \le L_0(1+|B|)^p e^{\theta|B|^2}, \qquad \|F_1(B) - F_0(B)\| \le L_\Delta(1+|B|)^{p_\Delta} e^{\theta_\Delta|B|^2}, \tag{UW12.21}$$

where A_0 is independent of this reference coordinate only. The old-state dependence of every displayed quantity is retained. Take the exponents so that adding the comparison budget in UW12.15 still lies inside the one available Gaussian/energy margin. One does not allocate a fresh copy of that margin to each summand or source word.

With $h = dK_1/dK_0$, the exact comparison is

$$K_1F_1 - K_0F_0 = K_1(F_1 - F_0) + K_0[(h-1)(F_0 - A_0)]. \tag{UW12.22}$$

Using UW12.16 with the actual polynomial/exponential envelopes gives

$$\|K_1F_1 - K_0F_0\| \le C_{M,p_\Delta,\theta_\Delta} L_\Delta + C_{M,p,\theta}\delta L_0. \tag{UW12.23}$$

The first term is not removed merely because the reference kernel is normalized. An independent old factor $e^{s(b)}$ multiplies the corresponding envelopes. Their degrees and old original meshes are unchanged; no degree-uniform moment constant is asserted.

If the scalar difference itself is being transported, one must use the raw identity in UW12.4, with the original coordinate factor and its genuine action-completion partner. A normalized comparison such as UW12.23 is not an estimate for an independently retained scalar mass.

158.6 Combining different weights with the original forest cutoff mismatch

The comparison can be combined with RV11's actual relative-forest transport when the fixed physical target map is the same and the only additional difference is its spatial weight. Use its transported physical cutoff domains Ω_0, Ω_1, and let $C = \Omega_0 \cap \Omega_1$. Suppose each normalized endpoint has exit mass $\tau_i = K_i(C^c) < 1/2$. On the common core the physical density ratio is exactly

$$d\mu_1 = e^{-c-w}\, d\mu_0, \qquad c = g^{-2}A(\Delta\zeta, U_0). \tag{UW12.24}$$

Put $K_{0,C} = K_0(\,\cdot \mid C)$. Directly comparing core masses gives

$$\log \frac{D_1}{D_0} = -c + \log K_{0,C}(e^{-w}) + \log(1-\tau_0) - \log(1-\tau_1). \tag{UW12.25}$$

The conditional moments are bounded by $K_{0,C}H \le K_0 H/(1-\tau_0)$ for nonnegative H. Consequently

$$\left|\log \frac{D_1}{D_0} + c\right| \le C_M\delta + 2(\tau_0+\tau_1). \tag{UW12.26}$$

The RV11 carrier identity and UW12.22 apply on the common core; the two exiting complete carriers remain bounded by their own weighted tails. If both endpoint native tails have been established, their rate is $C_M e^{-c_M p_0(g)^2}$. This is an additional cutoff tail, not a substitute for the e^{-cD} physical-distance term.

This statement does not transport a changed pin set, a changed number of free coordinates, or a different physical fixed-target map. For a forest change on one graph with the same pins, the original count is $n = m(|E| - |V| + |P|)$ and remains equal. If those data change, the raw coordinate-scalar difference must instead be kept and estimated; the core identification used above has not been supplied.

159.4 The derivative bound uses a genuinely slower rate

In the original (190), an output point x lies in the native cell at $y \in A_j$, and the input coordinate is at $y' \in A_{j'}$. The potential and first-difference factors are respectively

$$\sigma_j^{-1}\sigma_{j'}^{-d}, \qquad \sigma_j^{-2}\sigma_{j'}^{-d}, \tag{NM12.11}$$

multiplied by

$$e^{-(\delta_0/8)d(y,y')}. \tag{NM12.12}$$

For a top-scale input, $\sigma_{j'} = 1$. Passing to a curvature L^2 norm on one actual own-scale cell gives exactly the factor $v_Q^{1/2}\sigma_Q^{-2}$ in NM12.6, with $c = \delta_0/8 = \beta/4$. It cannot be inserted into NM12.7 using (162) alone.

This is not repaired by saying that δ_0 can be made smaller. The sum (162) has been asserted at its displayed rate for the original geometry. Replacing that rate by a quarter of itself requires a new summation estimate on the same geometry. It is the conclusion to prove, not a consequence of weakening a pointwise decay estimate.

159.5 Why the implication needs more than volume counting

There is a simple numerical illustration of the missing inference. It is **not** a counterexample to the native geometry. Set

$$\sigma_j = L^{-j}, \qquad N_j = L^{3j}, \qquad d_j = 5j \log L/\beta. \tag{NM12.13}$$

The total own-cell four-volume is finite:

$$\sum_j N_j \sigma_j^4 = \sum_j L^{-j} < \infty. \tag{NM12.14}$$

The faster weighted sum has the structure required by (162):

$$\sum_j N_j (d_j + 1)\sigma_j^{-1} e^{-\beta d_j} = \sum_j (d_j + 1)L^{-j} < \infty. \tag{NM12.15}$$

Nevertheless the desired slower unweighted sum diverges:

$$\sum_j N_j e^{-(\beta/4)d_j} = \sum_j L^{7j/4} = \infty. \tag{NM12.16}$$

Thus even the actual volume weight, the faster weighted estimate, and a surface-like own-scale cell count do not logically imply the slower sum. A stronger property of the native metric is needed.